



IT3021 – Data Warehousing & Business Intelligence

Assignment 02

IT Number – IT23629608
Name – Wijethilaka P.L.A.W.K
Module – IT3021

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Step 1 - Data Source

Overview

The data source used for Assignment 2 is the data warehouse developed in Assignment 1, named CarSales_DW. This was implemented in SQL Server and populated through ETL processes. It is designed using a star schema to support analytical processing and business intelligence reporting. The warehouse serves as the source for the SSAS cube, Excel OLAP analysis, and Power BI reports.

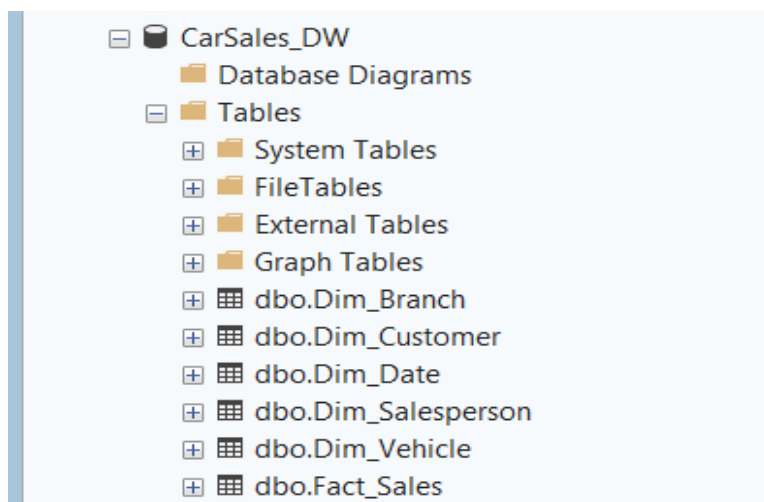


Figure 1: Tables in CarSales_DW database showing fact and dimension tables

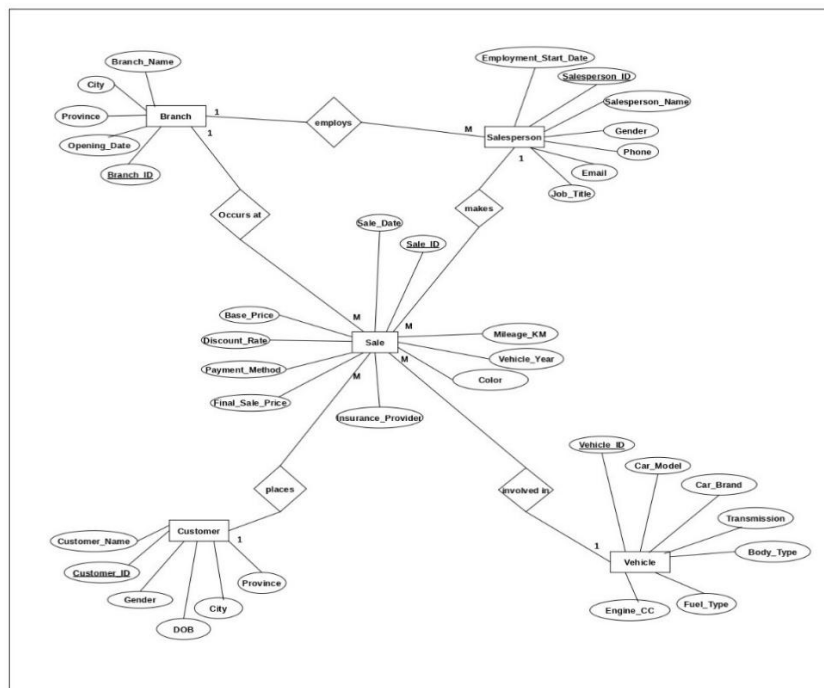


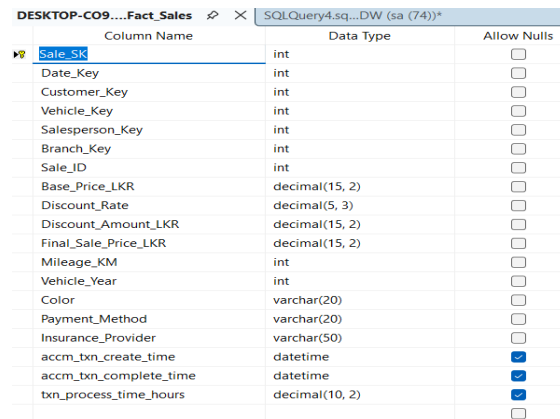
Figure 2: ER Diagram

Structure of the Data Warehouse

The warehouse consists of one central fact table and several dimension tables.

- **Fact_Sales**

Stores measurable transaction data related to vehicle sales, such as base price, discount rate, discount amount, final sale price, mileage, and vehicle year.

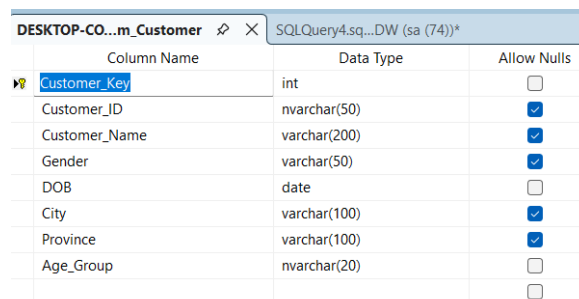


The screenshot shows the 'Fact_Sales' table structure in SQL Server Enterprise Manager. The table has 21 columns. The first 10 columns are integer keys, and the remaining 11 columns are decimal or datetime values. The 'Allow Nulls' column shows that most columns do not allow nulls, except for 'accm_txn_create_time', 'accm_txn_complete_time', and 'txn_process_time_hours' which do.

Column Name	Data Type	Allow Nulls
Sale_Sk	int	☐
Date_Key	int	☐
Customer_Key	int	☐
Vehicle_Key	int	☐
Salesperson_Key	int	☐
Branch_Key	int	☐
Sale_ID	int	☐
Base_Price_LKR	decimal(15, 2)	☐
Discount_Rate	decimal(5, 3)	☐
Discount_Amount_LKR	decimal(15, 2)	☐
Final_Sale_Price_LKR	decimal(15, 2)	☐
Mileage_KM	int	☐
Vehicle_Year	int	☐
Color	varchar(20)	☐
Payment_Method	varchar(20)	☐
Insurance_Provider	varchar(50)	☐
accm_txn_create_time	datetime	☒
accm_txn_complete_time	datetime	☒
txn_process_time_hours	decimal(10, 2)	☒

- **Dim_Customer**

Stores customer-related descriptive information such as customer ID, name, gender, city, province, and age group.



The screenshot shows the 'Dim_Customer' table structure in SQL Server Enterprise Manager. The table has 8 columns. The first column is an integer key, and the remaining 7 columns are string or date values. The 'Allow Nulls' column shows that 'Customer_ID', 'Customer_Name', 'Gender', 'City', and 'Province' do not allow nulls, while 'DOB' and 'Age_Group' do.

Column Name	Data Type	Allow Nulls
Customer_Key	int	☐
Customer_ID	nvarchar(50)	☒
Customer_Name	varchar(200)	☒
Gender	varchar(50)	☒
DOB	date	☐
City	varchar(100)	☒
Province	varchar(100)	☒
Age_Group	nvarchar(20)	☐

- **Dim_Vehicle**

Stores vehicle-related information such as vehicle ID, brand, model, engine capacity, fuel type, transmission, and body type.

Column Name	Data Type	Allow Nulls
Vehicle_Key	int	☐
Vehicle_ID	nvarchar(50)	☒
Car_Brand	varchar(50)	☒
Car_Model	varchar(50)	☒
Engine_CC	int	☐
Fuel_Type	varchar(50)	☒
Transmission	varchar(50)	☒
Body_Type	varchar(50)	☒
		☐

- **Dim_Branch**

Stores branch details such as branch ID, branch name, city, and province.

Column Name	Data Type	Allow Nulls
Branch_Key	int	☐
Branch_ID	nvarchar(50)	☒
Branch_Name	varchar(200)	☒
City	varchar(100)	☒
Province	varchar(100)	☒
Opening_Date	date	☐
		☐

- **Dim_Salesperson**

Stores salesperson-related details used for sales analysis.

Column Name	Data Type	Allow Nulls
Salesperson_Key	int	☐
Salesperson_ID	int	☐
Salesperson_Name	varchar(200)	☒
Gender	varchar(50)	☐
Phone	varchar(50)	☒
Email	varchar(200)	☒
Home_Branch	varchar(200)	☒
Job_Title	varchar(100)	☒
Start_Date	date	☐
End_Date	date	☒
Is_Current	bit	☐
		☐

- **Dim_Date**

Stores date-related attributes such as full date, month, quarter, and year to support time-based analysis.

Column Name	Data Type	Allow Nulls
Date_Key	int	☐
Full_Date	date	☐
Day	int	☐
Month	int	☐
Month_Name	nvarchar(20)	☐
Quarter	nvarchar(5)	☐
Year	int	☐
Day_of_Week	nvarchar(15)	☐
Is_Weekend	bit	☐

1.3 Data Warehouse Design

The data warehouse follows a **star schema design**, where Fact_Sales is the central fact table and all dimension tables are linked to it through surrogate keys. This design improves query performance and supports multidimensional analysis.

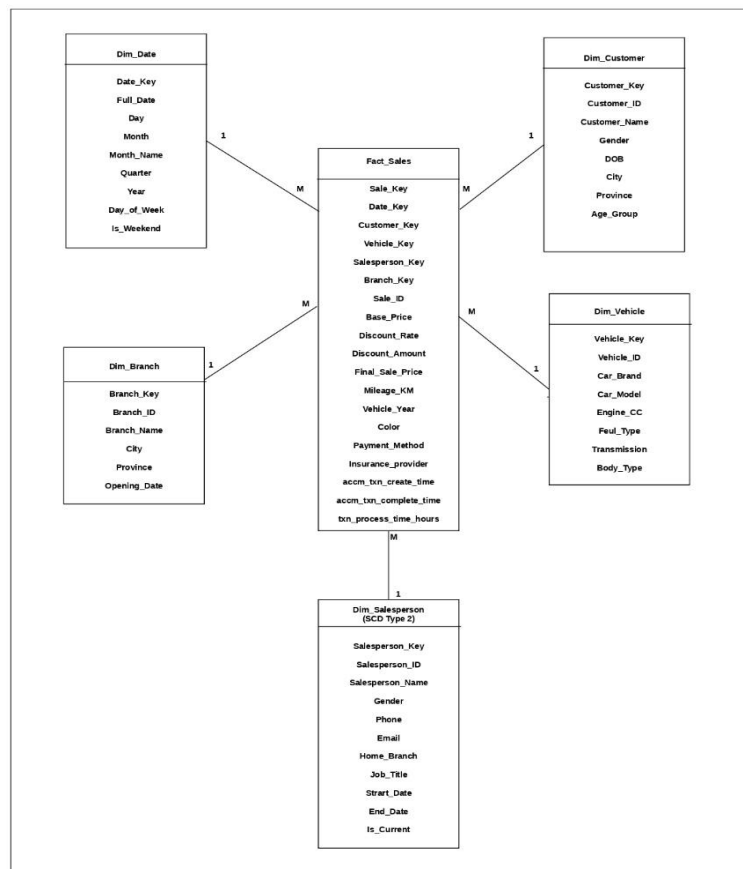
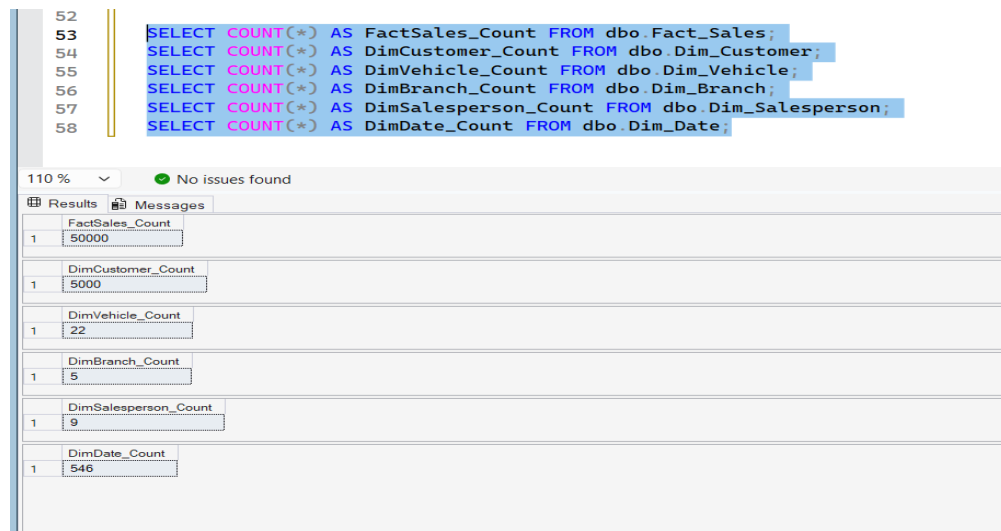


Figure 3: Star schema design of the CarSales_DW data warehouse

1.4 Suitability as a Data Source

This data warehouse is suitable as the data source for Assignment 2 because it contains integrated, structured, and analysis-ready data. The fact and dimension model supports the creation of cubes, hierarchies, pivot-based OLAP analysis, and interactive dashboards.



```
52
53 SELECT COUNT(*) AS FactSales_Count FROM dbo.Fact_Sales;
54 SELECT COUNT(*) AS DimCustomer_Count FROM dbo.Dim_Customer;
55 SELECT COUNT(*) AS DimVehicle_Count FROM dbo.Dim_Vehicle;
56 SELECT COUNT(*) AS DimBranch_Count FROM dbo.Dim_Branch;
57 SELECT COUNT(*) AS DimSalesperson_Count FROM dbo.Dim_Salesperson;
58 SELECT COUNT(*) AS DimDate_Count FROM dbo.Dim_Date;
```

110 % No issues found

Results	Messages
FactSales_Count	
1 50000	
DimCustomer_Count	
1 5000	
DimVehicle_Count	
1 22	
DimBranch_Count	
1 5	
DimSalesperson_Count	
1 9	
DimDate_Count	
1 546	

The tables were verified to be populated with data before being used as the source for Assignment 2.

Step 2 – SSAS Cube Implementation

2.1 Creating the SSAS Project

The SSAS project was created using Visual Studio to build the analytical cube for the data warehouse. A new project was initiated by selecting the Analysis Services Multidimensional Project option, and it was named "CarSalesCube" to reflect the dataset used. This project serves as the foundation for designing the cube, including defining measures, dimensions, and hierarchies required for analysis.

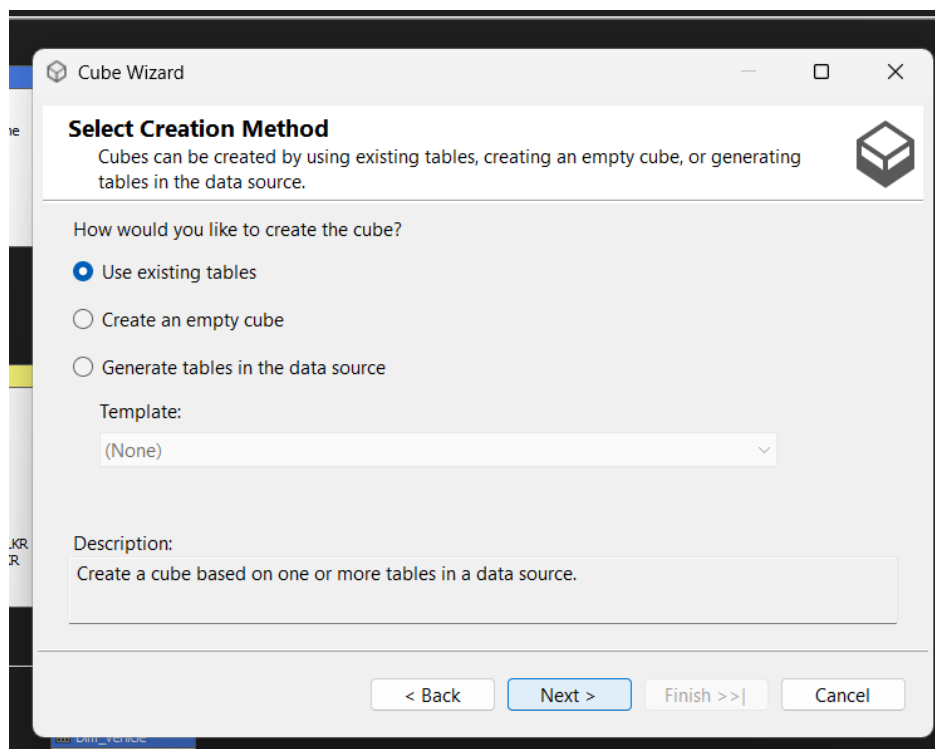


Figure 4: Creating Analysis Services Multidimensional Project in Visual Studio

2.2 Data Source

I created a data source to establish a connection between the SSAS cube and the CarSales_DW database. This was done by adding a new data source in the project and selecting the appropriate SQL Server instance. The CarSales_DW database was then chosen as the source of data, and Windows Authentication was used to ensure a secure connection. This step allows the cube to access the required tables for analysis.

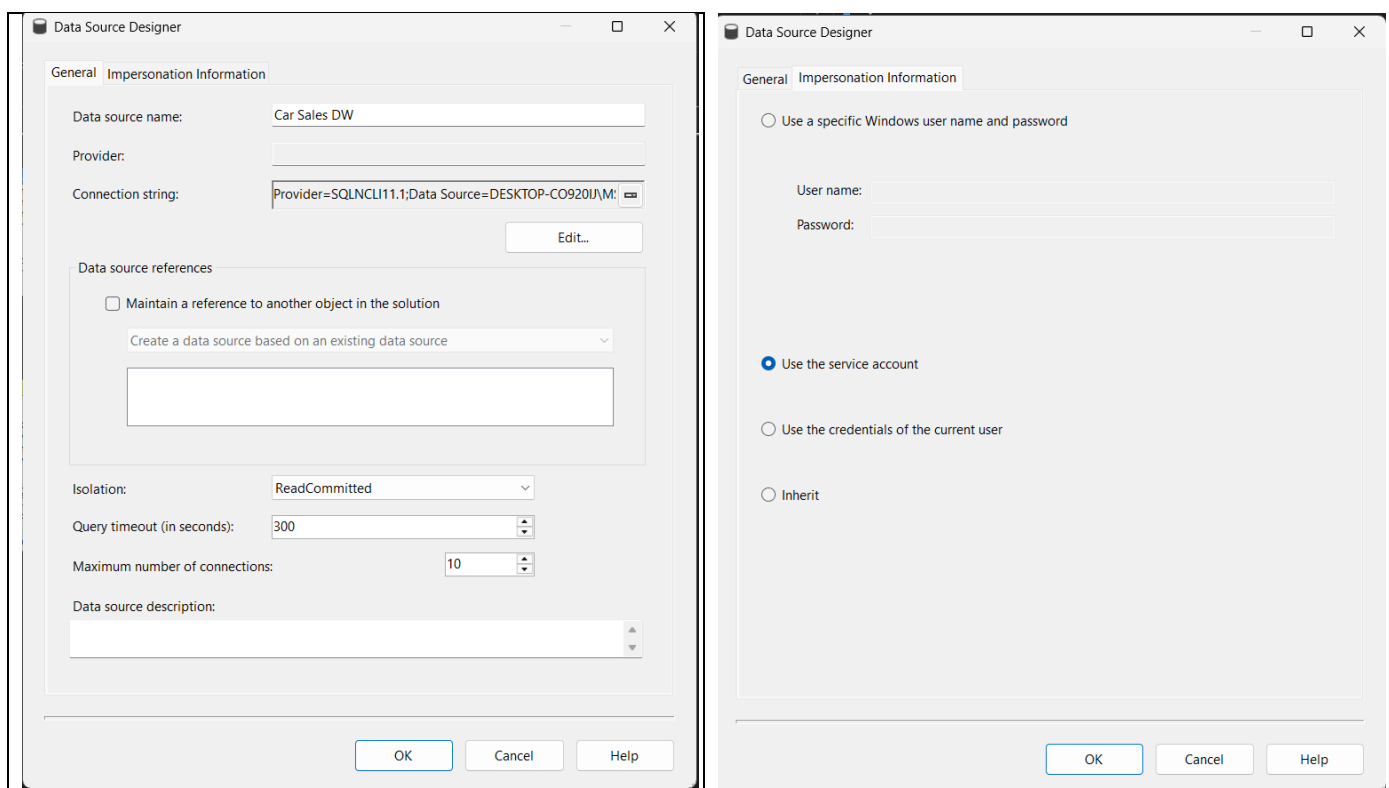


Figure 5:Configuring data source connection to CarSales_DW database

2.3 Data Source View

A Data Source View (DSV) was created to represent the structure of the CarSales_DW database within the SSAS project. This view includes the main Fact_Sales table along with all related dimension tables such as Dim_Date, Dim_Customer, Dim_Vehicle, Dim_Branch, and Dim_Salesperson. These tables were added to provide both transactional and descriptive data required for analysis.

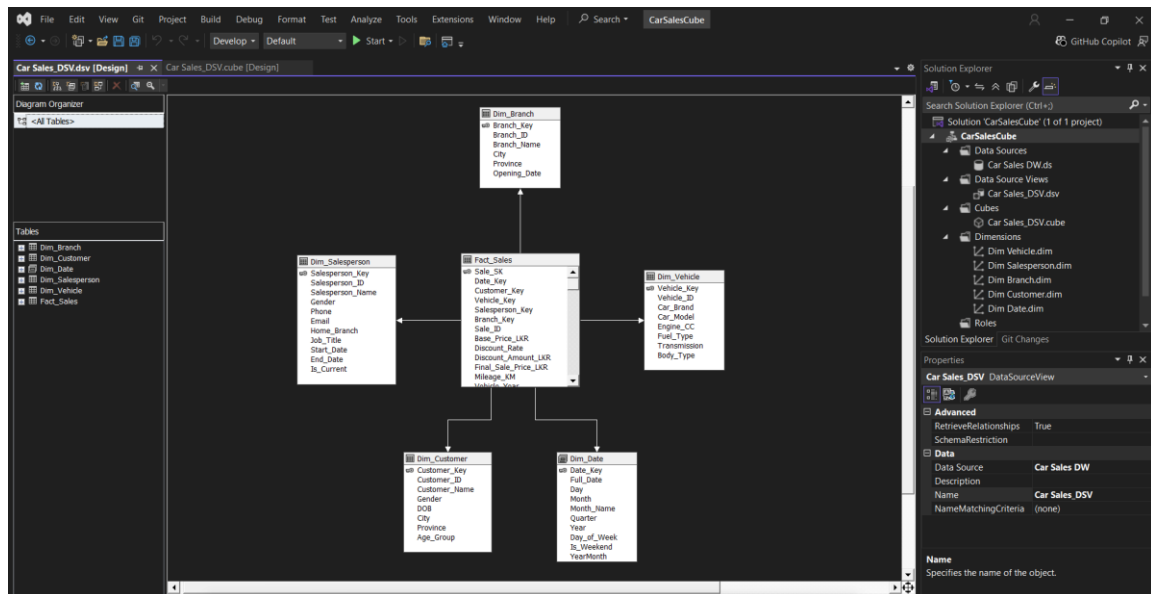


Figure 6: Data Source View showing fact and dimension tables with relationships

2.4 Designing the Cube

The cube was designed by selecting the Fact_Sales table as the main source of numerical data, which acts as the measure group. Important measures such as Final Sale Price, Base Price, Discount Amount, and Sales Count were included to support business analysis. Relevant dimensions were then linked to the cube, including Date, Customer, Vehicle, Branch, and Salesperson. These dimensions provide different perspectives for analyzing the sales data.

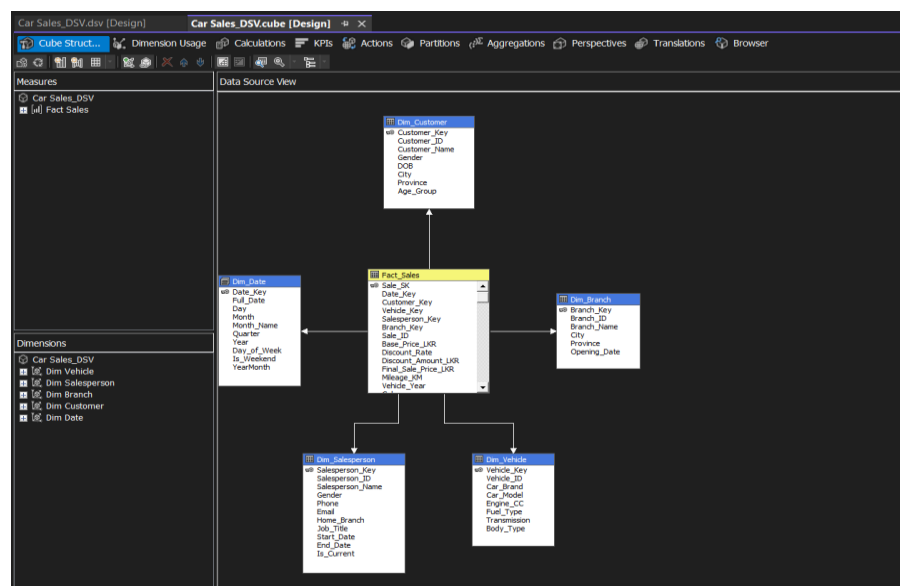


Figure 7: Cube wizard showing selection of measure group and dimensions

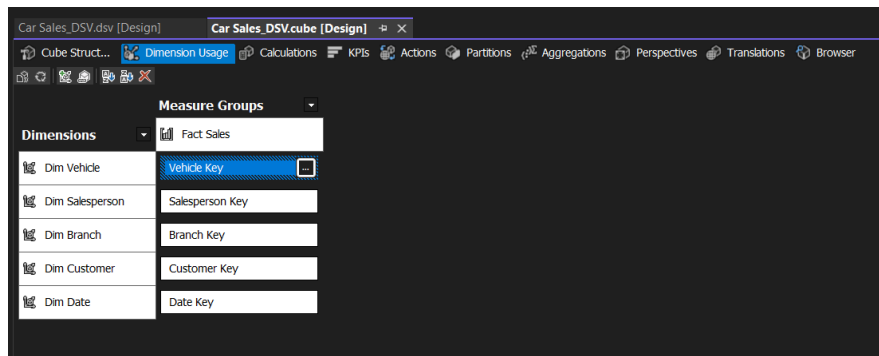


Figure 8:Cube structure displaying measures and dimensions

The fact table is connected to dimension tables through foreign key relationships, enabling proper data analysis.

2.5 Implementing Hierarchies

Hierarchies were implemented in the data warehouse to support structured data analysis and improve query performance in OLAP operations. These hierarchies allow users to drill down from higher-level aggregated data to more detailed levels.

In the **Dim_Date** dimension, a time hierarchy was created using the attributes:

- Year → Quarter → Month → Day

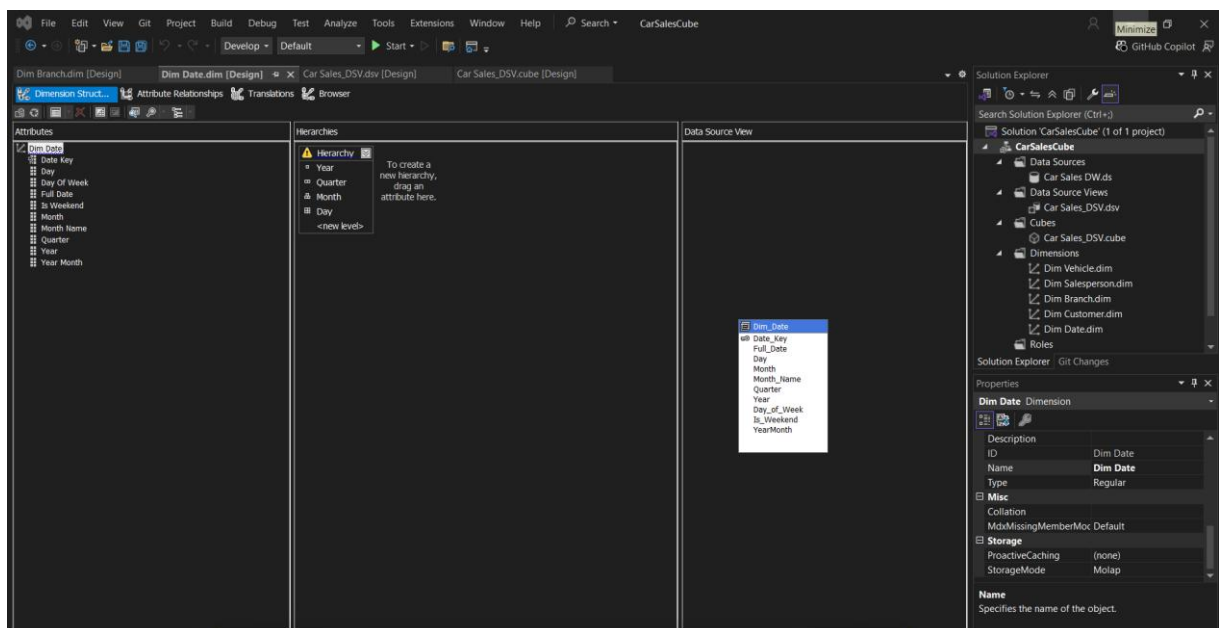


Figure 9:Date hierarchy showing Year, Quarter, Month, and Day levels

This hierarchy enables users to analyze sales data at different time levels, such as yearly trends, quarterly comparisons, monthly performance, and daily transactions. It supports drill-down and roll-up operations in analytical tools like Power BI.

In the **Dim_Branch** dimension, a geographical hierarchy was created using:

- Province → City → Branch Name

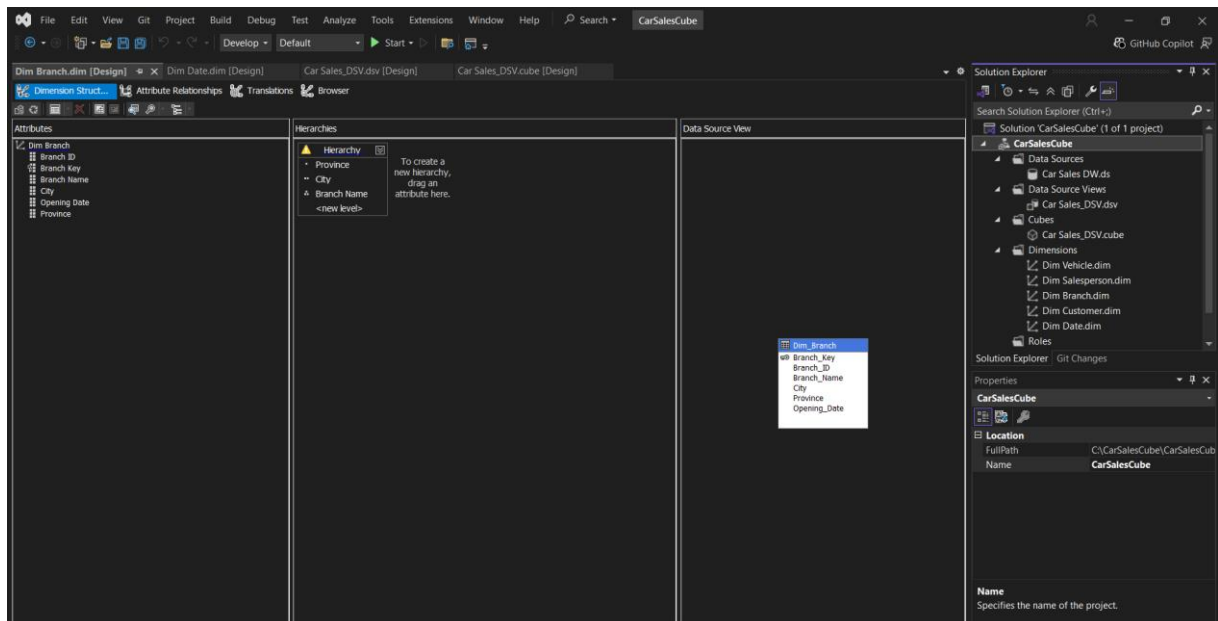


Figure 10: Branch hierarchy showing Province, City, and Branch Name levels

This hierarchy allows users to explore sales performance across different regions, starting from province level down to specific branches.

These hierarchies improve the usability of the cube by organizing data into meaningful structures and enhancing the efficiency of query execution in Analysis Services.

2.6 Deployment and Processing

After completing the cube design, the project was deployed to the Analysis Services server so that it can be accessed for analysis and reporting.

Once deployed, the cube was processed to load data from the data warehouse into the cube. This step prepares the cube by organizing the data into dimensions and measures.

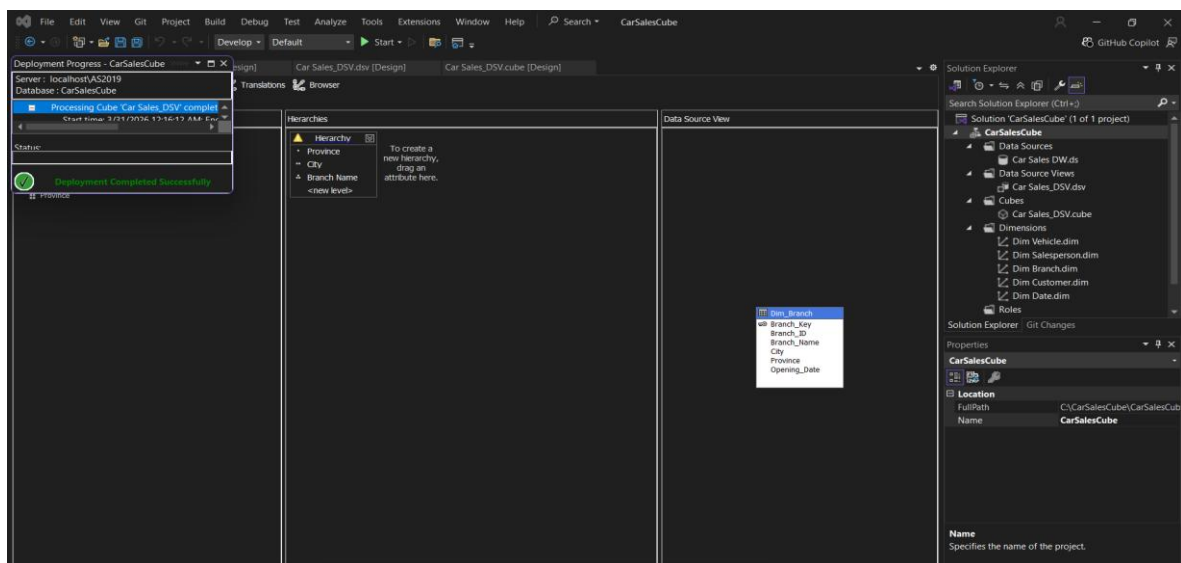


Figure 11: Successful deployment

After successful processing, the cube becomes ready to be used in tools such as Excel and Power BI for further analysis and visualization.

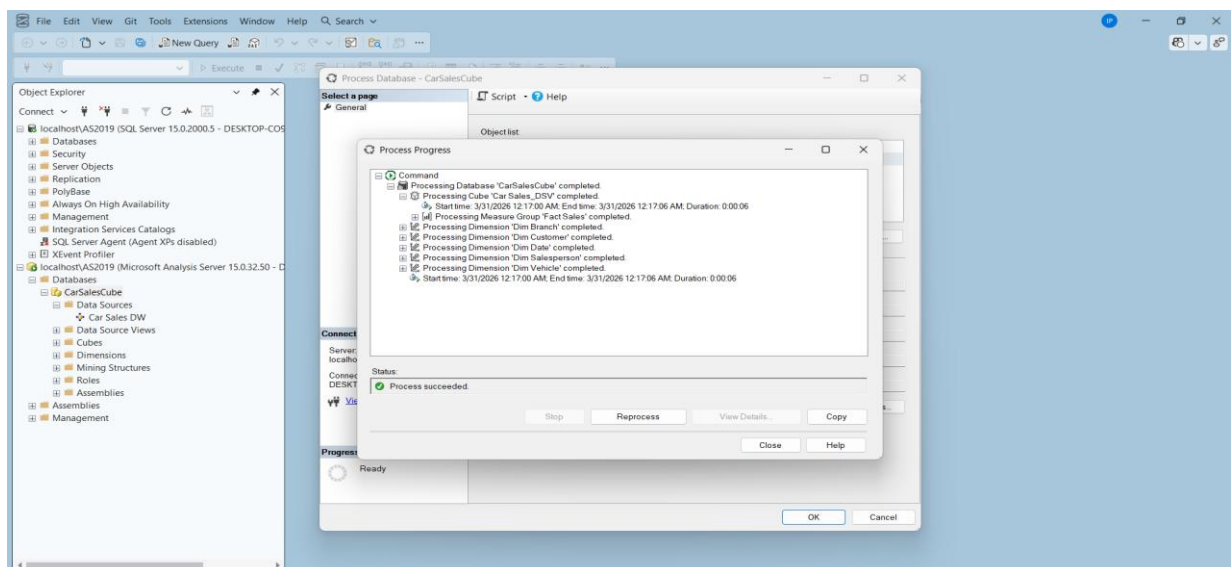


Figure 12: Successful processing of the SSAS cube

Step 3 – Demonstration of OLAP Operations

3.1 Connecting Excel to SSAS Cube

Microsoft Excel was connected to the SSAS cube to perform OLAP analysis. This was done using the “From Analysis Services” option in Excel, where the server name and cube were selected. Once connected, the cube data was loaded into a PivotTable, allowing interactive analysis using dimensions and measures.

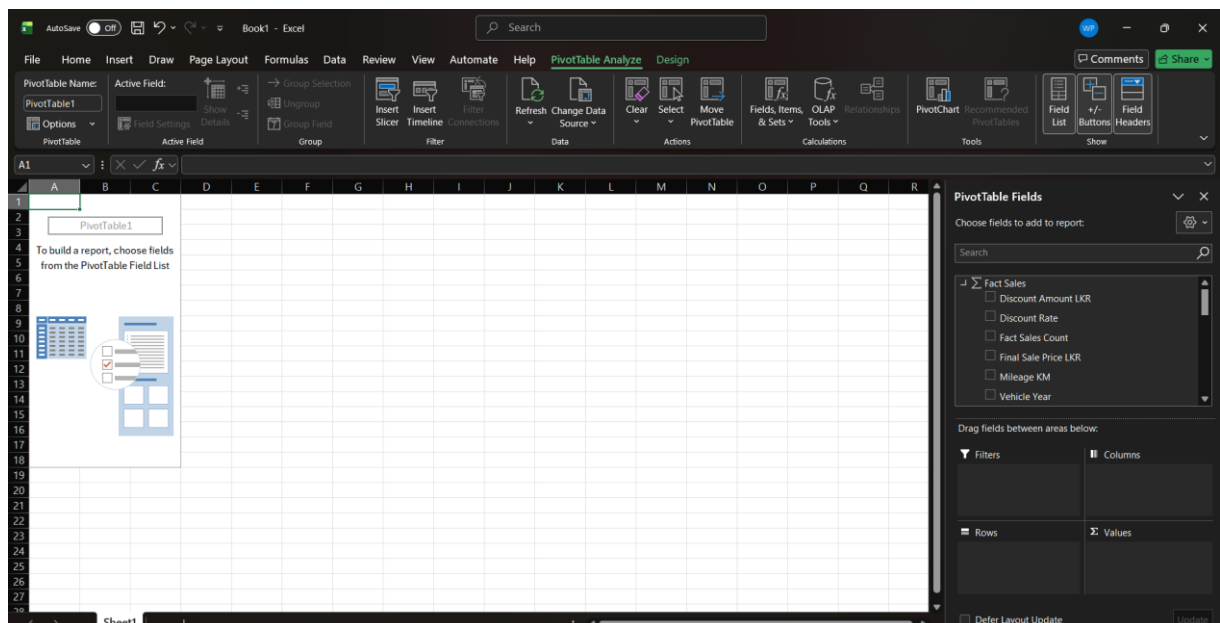
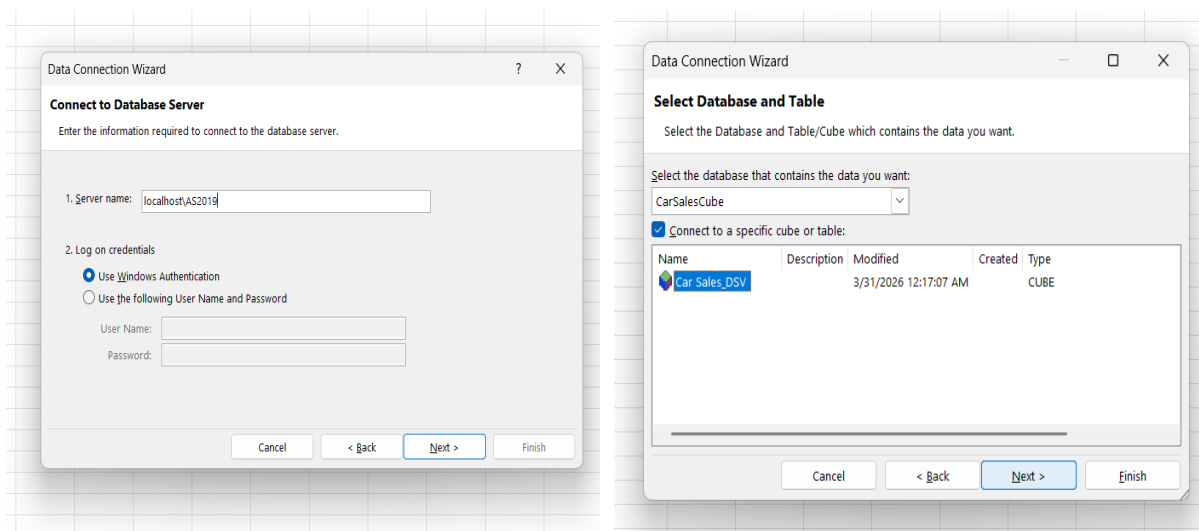


Figure 13: Connecting Excel to SSAS cube using Analysis Services

3.2 Roll-Up Operation

The roll-up operation was carried out to aggregate the data at a higher level.

In this analysis, the sales data was aggregated at the province level based on the hierarchy defined in the cube. The pivot table shows the total data for the Final Sale Price for each province, including Central, Northern, Southern, and Western.

A chart was created to show the data at the province level in terms of the total sales made. This helps in visualizing the data at a higher level without going into the details.

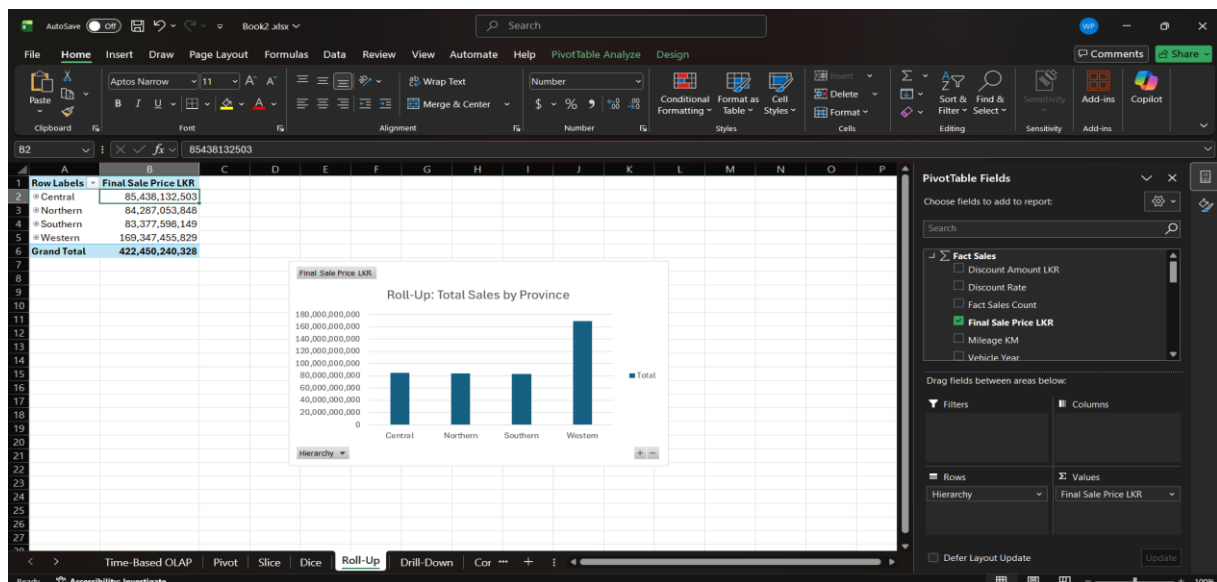


Figure 14: Roll-up operation showing aggregated sales by province

3.3 Drill-Down Operation

The drill-down operation was used to transition from the summary data to the more detailed data.

For this problem, the analysis began with the province level and then proceeded to the more detailed level. The pivot table indicates the data hierarchy: Province -> City -> Branch Name. This offers a better perspective on the sales data.

The chart offers a visual comparison of the sales data in the various branches in the region. This helps in identifying the branch that contributes the most to the sales.

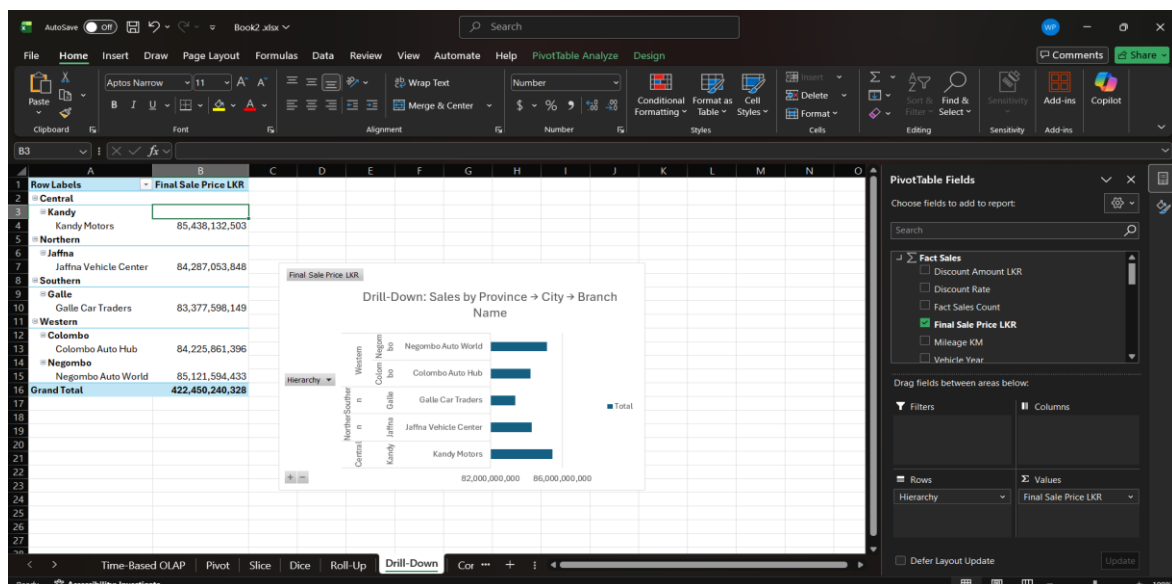


Figure 15: Drill-down operation from province to branch level

3.4 Slice Operation

The slice operation was used to filter the dataset based on a single dimension.

A slicer was applied on the Province field, and a specific value (Western) was selected. This allows the report to display only the sales data related to that selected province.

The resulting chart shows sales distribution within the selected region, making it easier to focus on a particular area without interference from other data.

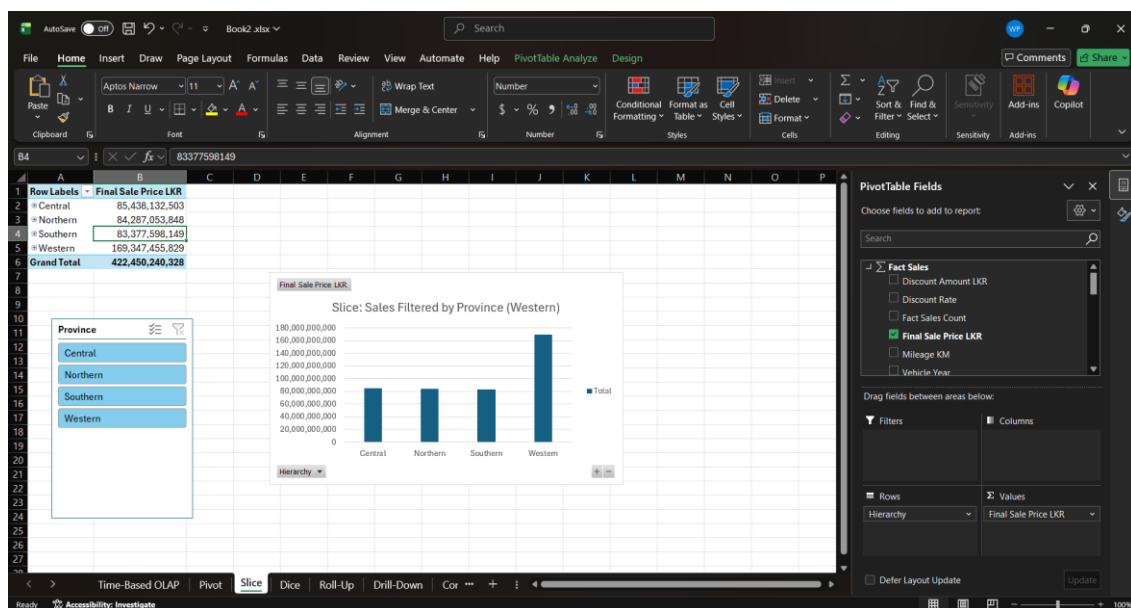


Figure 16: Slice operation filtering data by selected province

3.5 Dice Operation

The dice operation was carried out by making filters on multiple dimensions at the same time.

In the above analysis, Province and Age Group were used as filters. For instance, if we make the selection on the province as Western and age group as 25-34 years, it will be possible to view the data in more detail.

The chart displays sales values for the selected combination, enabling more detailed and targeted analysis based on multiple conditions.

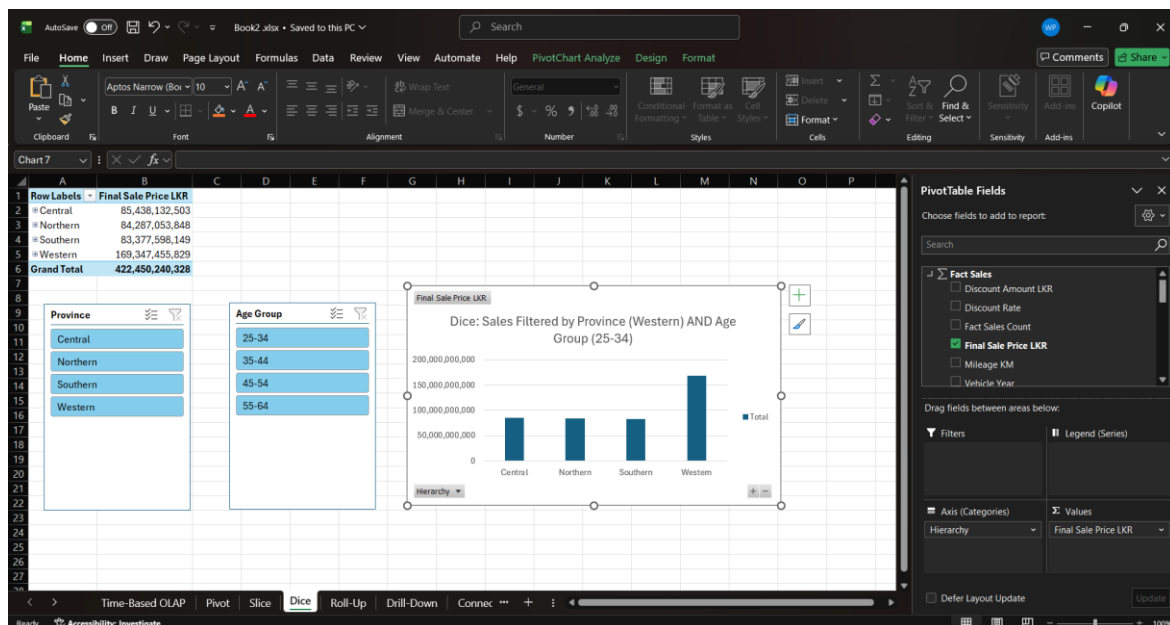


Figure 17: Dice operation using multiple filters (province and age group)

3.6 Pivot Operation

The use of the pivot operation was to rearrange the data in order to look at the data in a different perspective.

In the report, the sales data was organized according to the Age Group and the Branch Name. This is to enable the comparison of the sales at different age groups in each branch.

The pivot chart is helpful in understanding the contribution of different age groups of customers to the sales in the different branches.

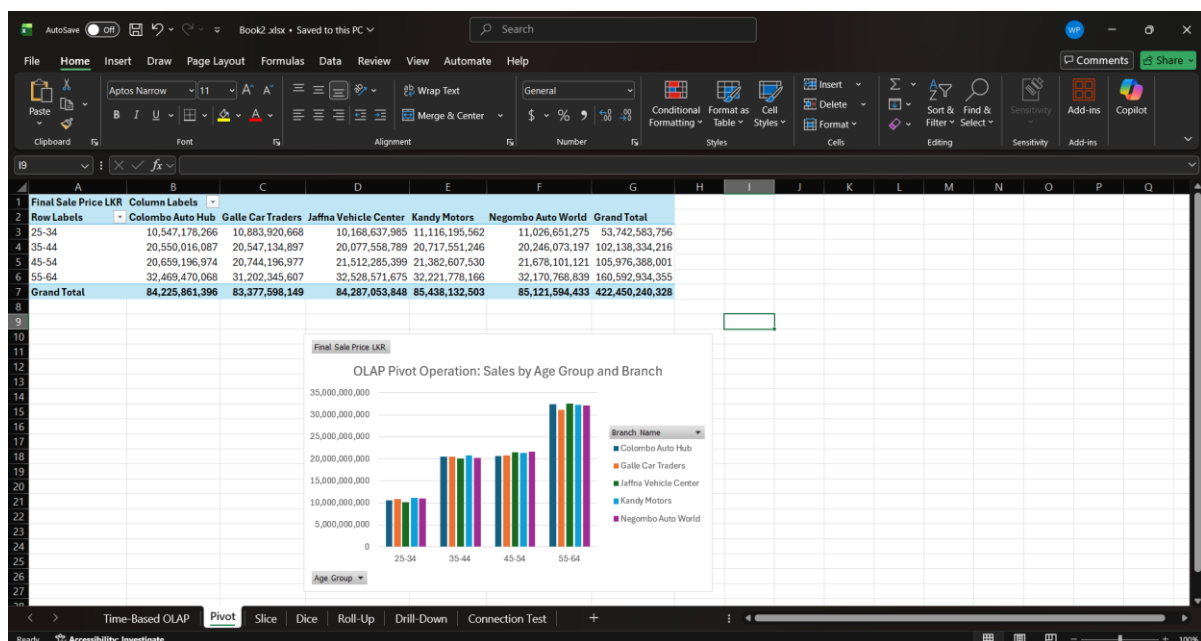


Figure 18: Pivot operation rearranging data by age group and branch

3.7 Time-Based Analysis

A time-based analysis was also performed using the date hierarchy in the cube.

Sales data was analyzed across different years, allowing comparison between 2022 and 2023. The chart shows how sales values vary over time, helping to identify trends and changes in performance.

This type of analysis is useful for understanding business growth and seasonal patterns.

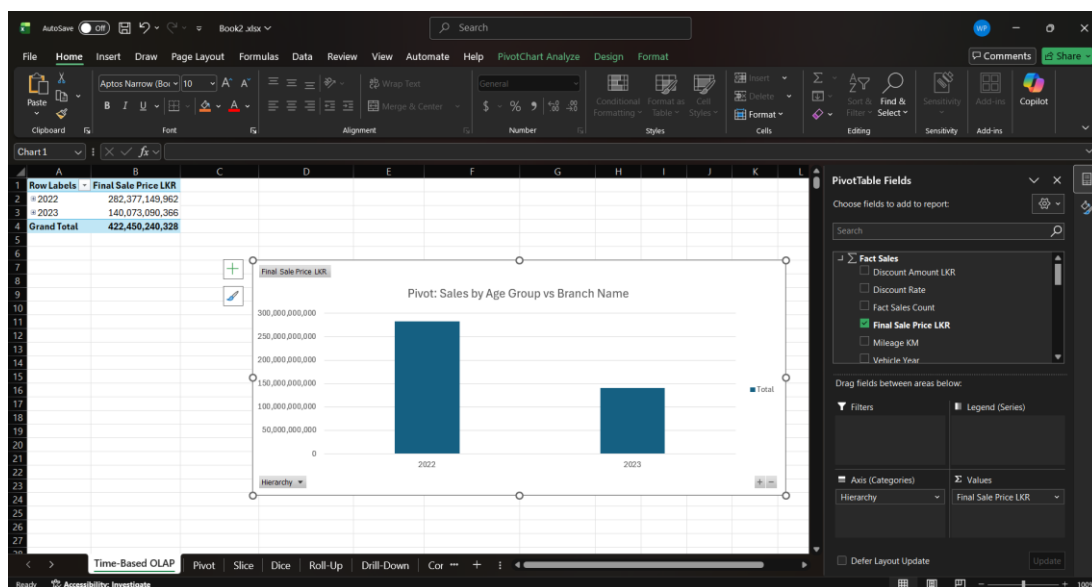


Figure 19: Time-based analysis showing sales trends across years

Step 4 – Power BI Report

Power BI Desktop was connected to the CarSales_DW database using the SQL Server data source option. First, the “Get Data” feature was used to select SQL Server as the data source. The server name and database name were then provided, and the connection mode was set to Import to load the data into Power BI.

After establishing the connection, the Navigator window was used to preview and select the required tables, including Fact_Sales and all relevant dimension tables such as Dim_Customer, Dim_Date, Dim_Vehicle, Dim_Branch, and Dim_Salesperson. These tables were then loaded into Power BI for further analysis.

This connection allowed the prepared data warehouse to be directly used for building interactive reports and visualizations.

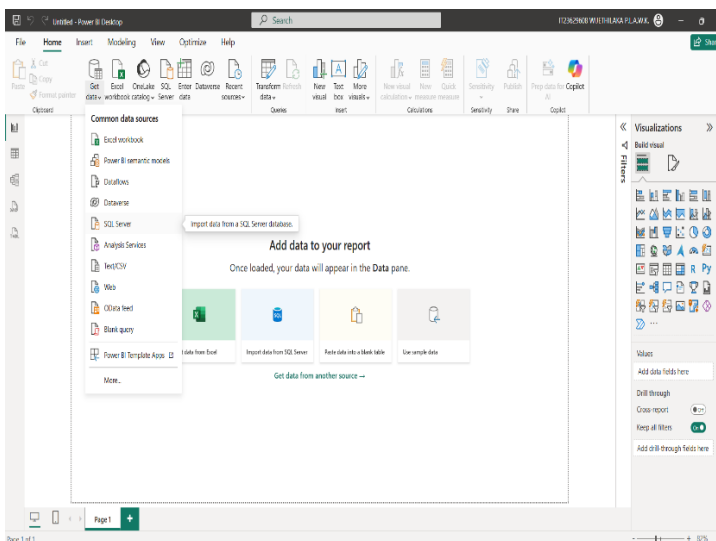


Figure 20: Connecting Power BI to SQL Server database (CarSales_DW)

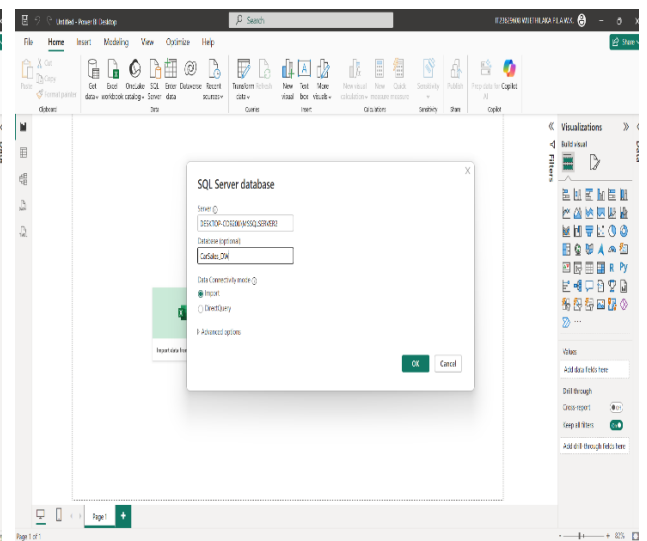


Figure 21: Navigator window showing selected tables for import

Navigator

Display Options

- DESKTOP-CO9201\MSQLSERVER2: CarSales_...
- Dim_Branch
- Dim_Customer
- Dim_Date
- Dim_Salesperson
- Dim_Vehicle
- Fact_Sales**
- Fact_Sales_Backup
- Lookup
- sysdiagrams
- fn_diagramobjects

Fact_Sales

Sale_SK	Date_Key	Customer_Key	Vehicle_Key	Salesperson_Key	Branch_Key
150001	20220301	27231	127	14	
150002	20230301	27232	112	10	
150003	20220422	27233	120	11	
150004	20220806	27234	120	10	
150005	20220704	27235	116	14	
150006	20230604	27236	119	14	
150007	20220306	27237	117	14	
150008	20220331	27238	127	13	
150009	20220813	27239	126	14	
150010	20220707	27240	129	15	
150011	20221025	27241	122	18	
150012	20230521	27242	115	11	
150013	20230522	27243	118	10	
150014	20220523	27244	120	13	
150015	20230429	27245	126	13	
150016	20230526	27246	117	12	
150017	20230111	27247	125	12	
150018	20220805	27248	127	10	
150019	20221204	27249	128	11	
150020	20220113	27250	127	16	

The data in the preview has been truncated due to size limits.

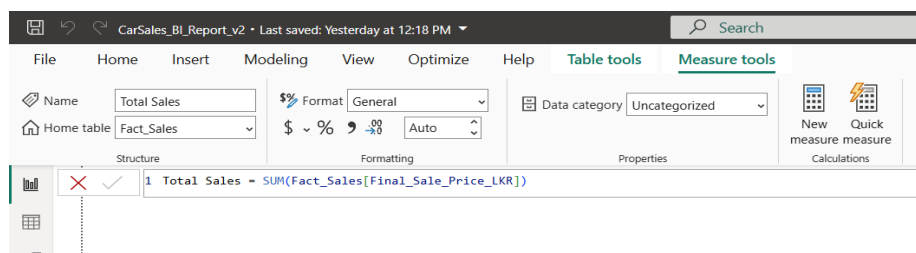
Select Related Tables Load Transform Data Cancel

Figure 22: Data loaded into Power BI for report creation

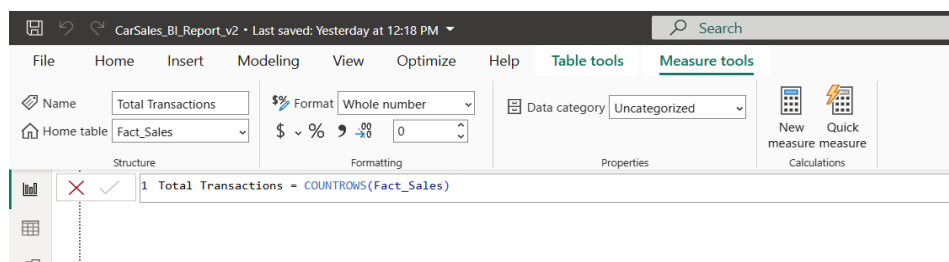
Prior to generating the reports, data modeling was done in Power BI to identify the relationships among the imported tables. This is where the table Fact_Sales was connected to each dimension table using the appropriate surrogate keys, namely Customer Key, Vehicle Key, Branch Key, Salesperson Key, and Date Key.

The following five DAX measures were created in the Fact_Sales table to support the reports and KPI Dashboard:

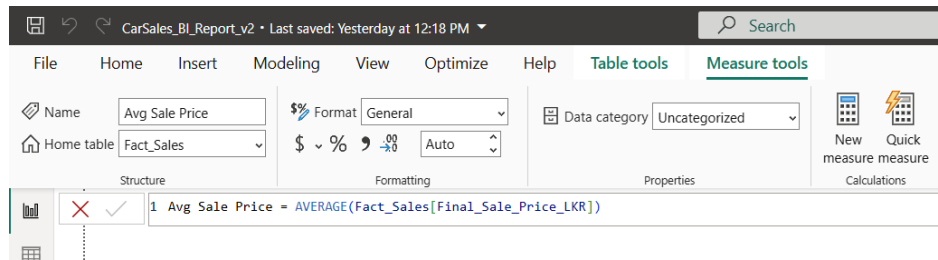
Total Sales — Calculates the sum of all final sale prices across all transactions. Total Sales = SUM(Fact_Sales[Final_Sale_Price_LKR])



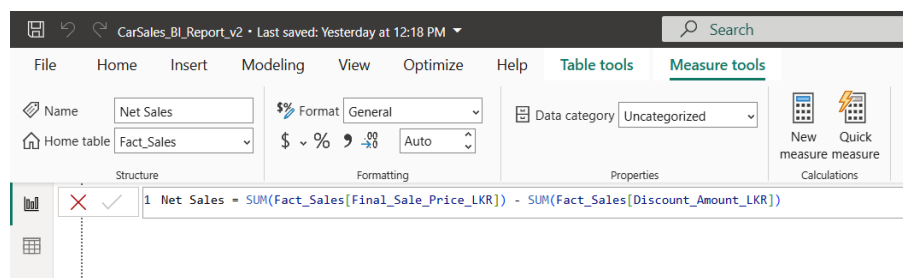
Total Transactions — Counts the total number of rows in the Fact_Sales table. The measure is formatted as a Whole Number. Total Transactions = COUNTROWS(Fact_Sales)



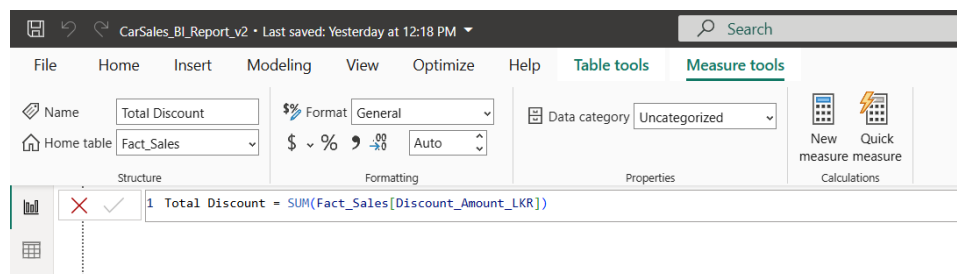
Avg Sale Price — Calculates the average final sale price per transaction. Avg Sale Price = $AVERAGE(Fact_Sales[Final_Sale_Price_LKR])$



Net Sales — Calculates revenue after deducting total discounts from total sales. Net Sales = $SUM(Fact_Sales[Final_Sale_Price_LKR]) - SUM(Fact_Sales[Discount_Amount_LKR])$



Total Discount — Calculates the total discount amount applied across all sales. Total Discount = $SUM(Fact_Sales[Discount_Amount_LKR])$



The above metrics were applied throughout all four documents and the KPI Dashboard, thus providing consistency within the data. No further data manipulation was necessary, aside from what had been done through the ETL process in Assignment 1.

4.1 Report 1 — Matrix Visual

A matrix visual has been designed to represent sales data in a structured and hierarchical manner. The report indicates total sales amounts, which are grouped by Province and Branch Name, while the data is represented in rows, with Customer Age Group represented in columns, allowing for multiple groupings in a single visual.

This type of visual is used to compare sales data in different regions, branches, and age groups simultaneously. The row and column totals are provided, which makes this visual suitable for both in-depth analysis and summarization.

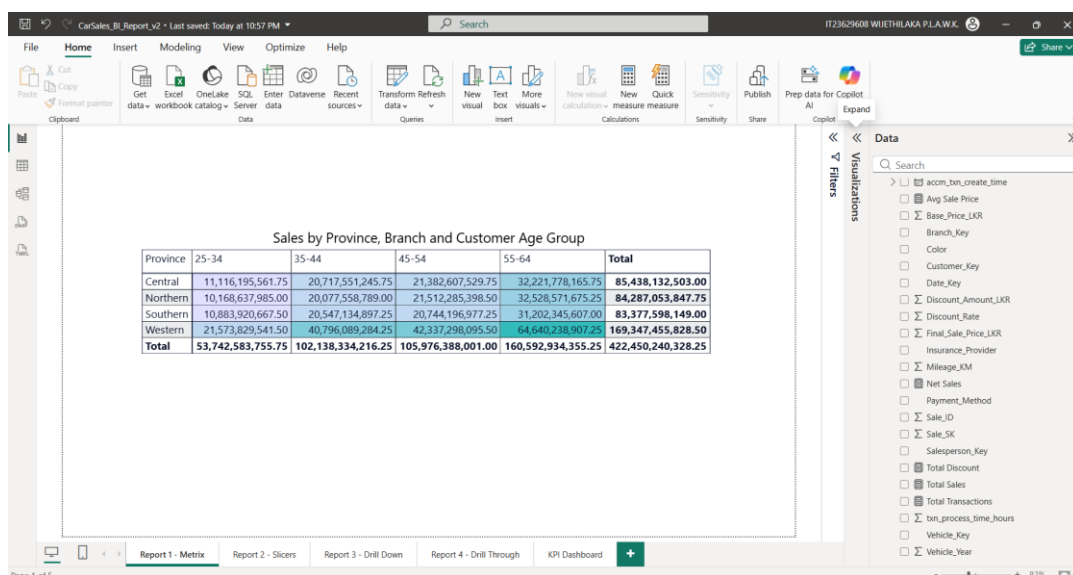


Figure 23: Matrix visual showing sales by province and age group

From the analysis, it is clear that the province with the highest total sales in terms of LKR 169.35bn is the Western province. In addition, the 55-64 age group has the highest purchasing power in all provinces compared to other age groups.

4.2 Report 2 — Slicers and Interactive Dashboard

An interactive dashboard was developed using multiple slicers and visualizations to enhance data exploration. Four slicers were added - **Year**, **Quarter**, **Province**, and **Payment Method**, allowing users to dynamically filter all visuals simultaneously.

The report includes a **Branch Performance Summary** table showing Total Sales, Total Transactions, and Average Sale Price per branch, a **bar chart** displaying Total Sales by Salesperson Name, and a **bar chart** showing Total Sales by Vehicle Color. Selecting any slicer value automatically filters the entire dashboard, providing focused and relevant analysis.

The slicers are interconnected, where selecting a value in one slicer dynamically filters the available options in the other slicers, implementing cascading filtering.

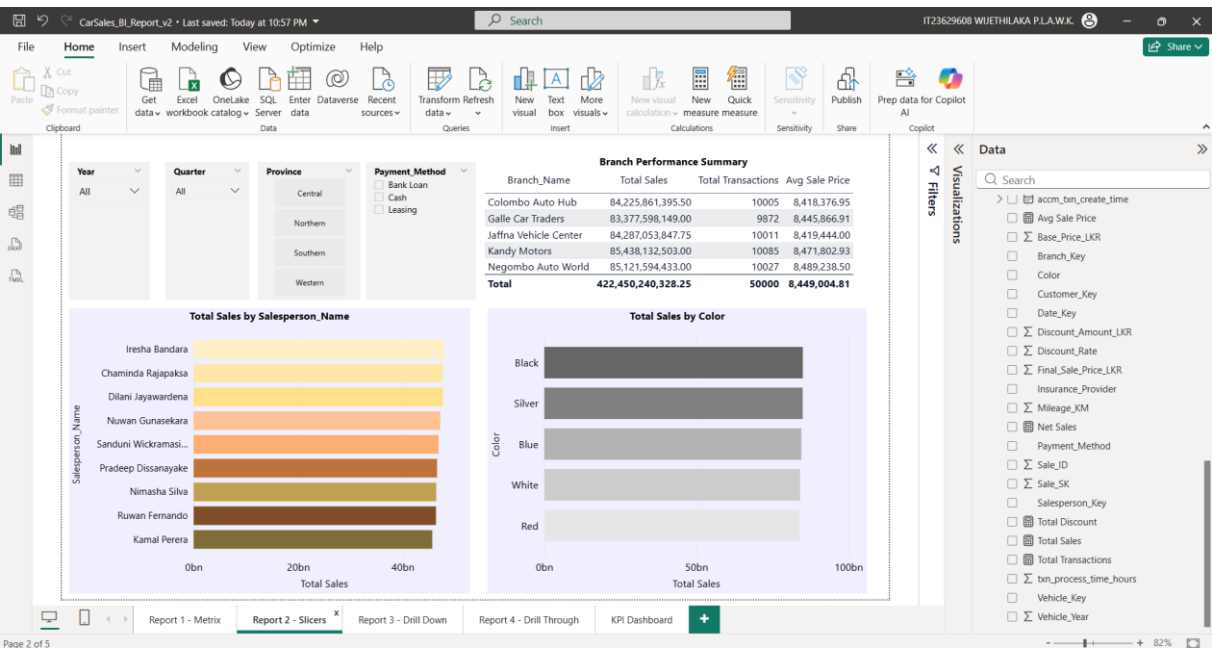


Figure 24: Interactive dashboard with slicers and multiple visuals

From the dashboard, it can be observed that all five branches perform comparably in total sales, each generating approximately LKR 83–85bn. Among salespersons, performance is relatively balanced, while Black and Silver colored vehicles generate the highest sales compared to other colors.

4.3 Report 3 — Drill-Down Report

In order to analyze sales trends at different levels of temporal and geographic granularity, a drill-down report is created.

Two drill-down visuals have been created for this purpose. The first is based on a hierarchical date structure of Year -> Quarter -> Month. This will enable the user to interact with the chart and drill from a yearly view to a quarterly view and further to a monthly view. The second is based on a geographic hierarchy of Province -> Branch Name. This will enable the user to drill from a provincial view to a branch view.

This feature will provide a better level of understanding of trends and patterns by allowing the user to understand how sales performance changes over time and across locations.

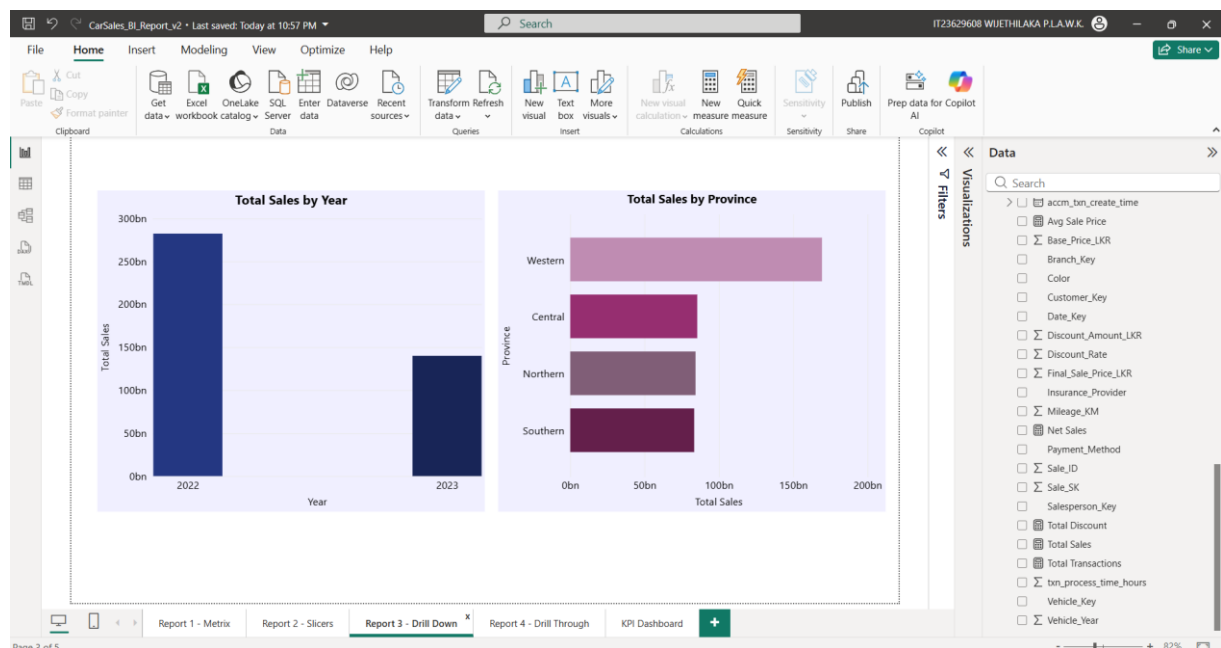
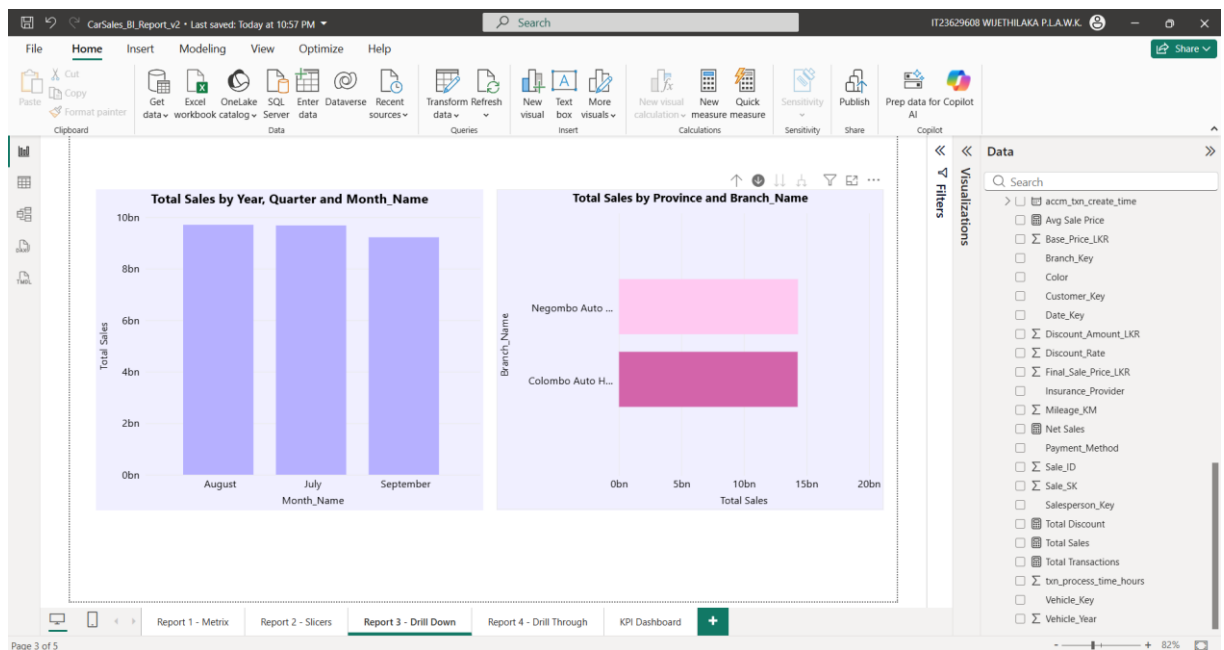
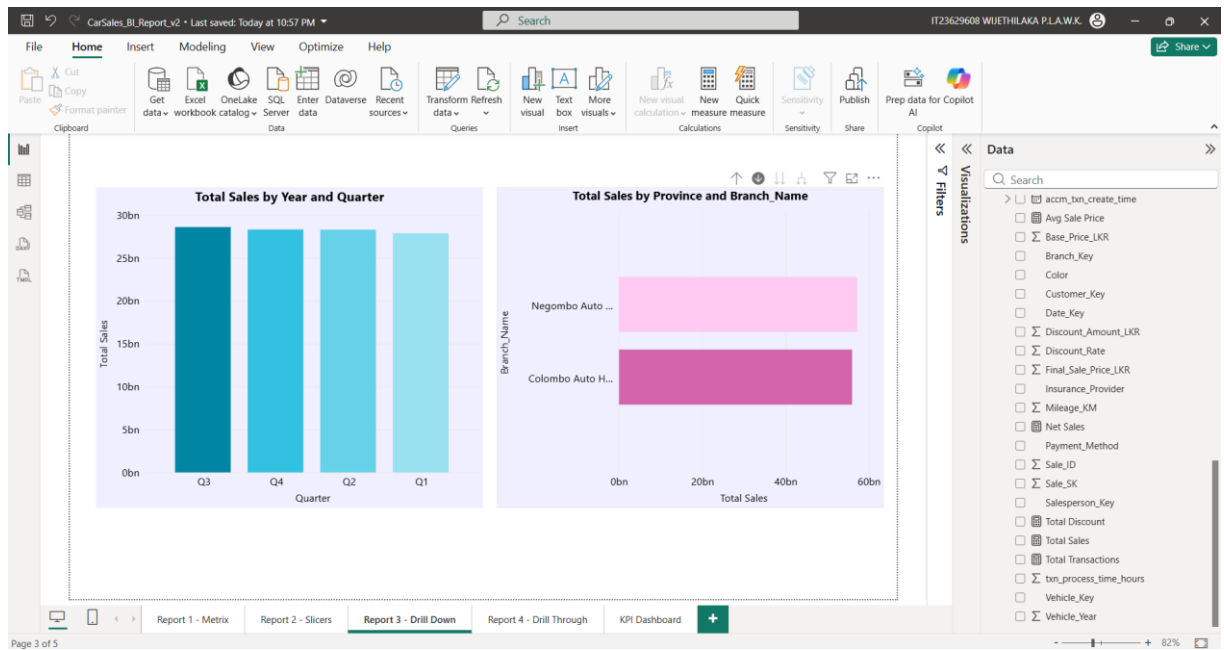


Figure 25: Drill-down report showing sales by year, quarter, and month



From the yearly comparison, sales in **2022 are significantly higher than 2023**, suggesting either a decline in performance or that 2023 data covers only a partial year. At the province level, **Western province** consistently leads across all drill levels.

4.4 Report 4 — Drill-Through Report

A drill-through report was implemented to provide detailed branch-level insights when navigating from other report pages.

Users can right-click on a Province data point in any other report page and select **"Drill through -> Report 4"** to be automatically redirected to this page with filters applied to the selected province. A **back navigation button** is automatically provided to return to the source page.

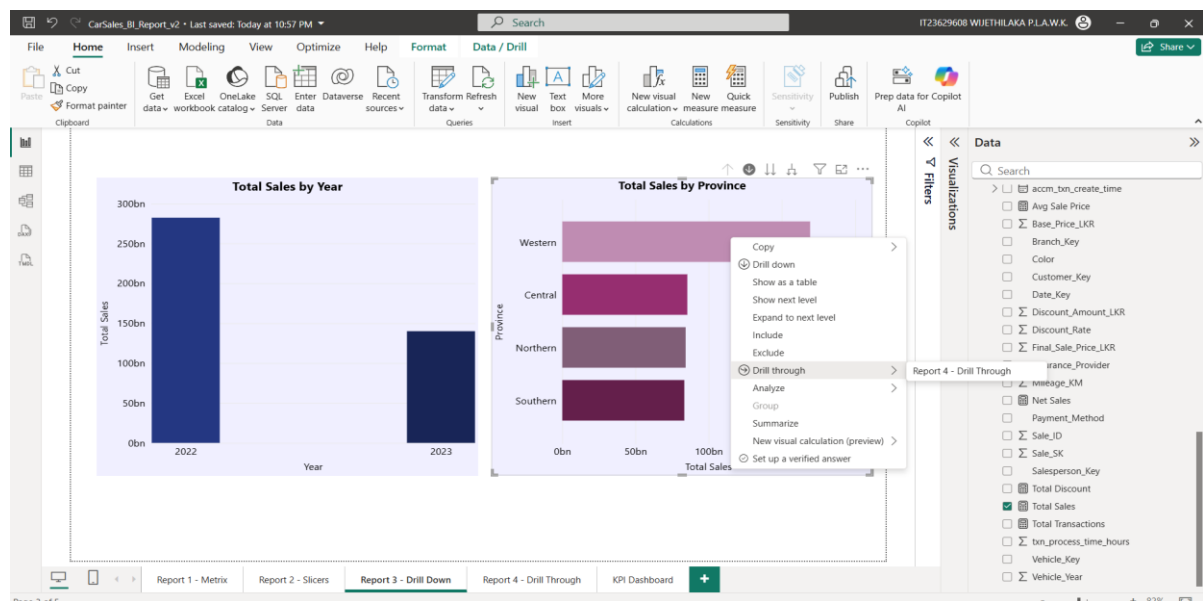
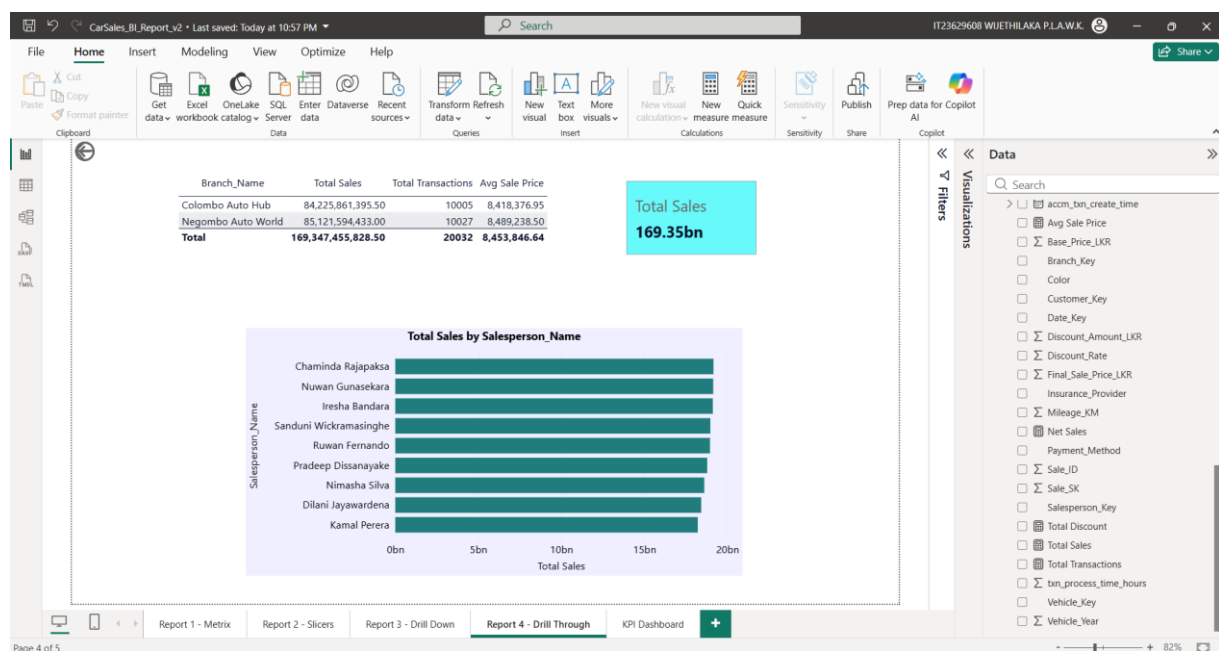


Figure 26: Drill-through report displaying detailed branch-level data



The report contains three visuals: a **Branch Performance table** displaying Branch Name, Total Sales, Total Transactions, and Average Sale Price; a **Total Sales KPI card** showing the filtered total; and a **bar chart** of Total Sales by Salesperson Name. All visuals respond dynamically to the drill-through filter, allowing users to focus on specific provincial data without manual filter adjustments.

4.5 KPI Dashboard

A KPI Dashboard was developed to provide a high-level executive summary of overall car sales performance.

The dashboard features four **KPI Cards** at the top displaying key metrics: **Total Sales (LKR 422.45bn)**, **Total Transactions (50K)**, **Average Sale Price (LKR 8.45M)**, and **Net Sales (LKR 404.83bn)**. Below the cards, a **horizontal bar chart** shows Total Sales by Branch Name for quick regional comparison, a **line chart** displays Total Sales trends by Month Name, and a **donut chart** illustrates the sales distribution by Payment Method across Cash, Leasing, and Bank Loan categories.

Two slicers - **Month Name** and **Province** - are provided to allow dynamic filtering of all visuals on the dashboard simultaneously.

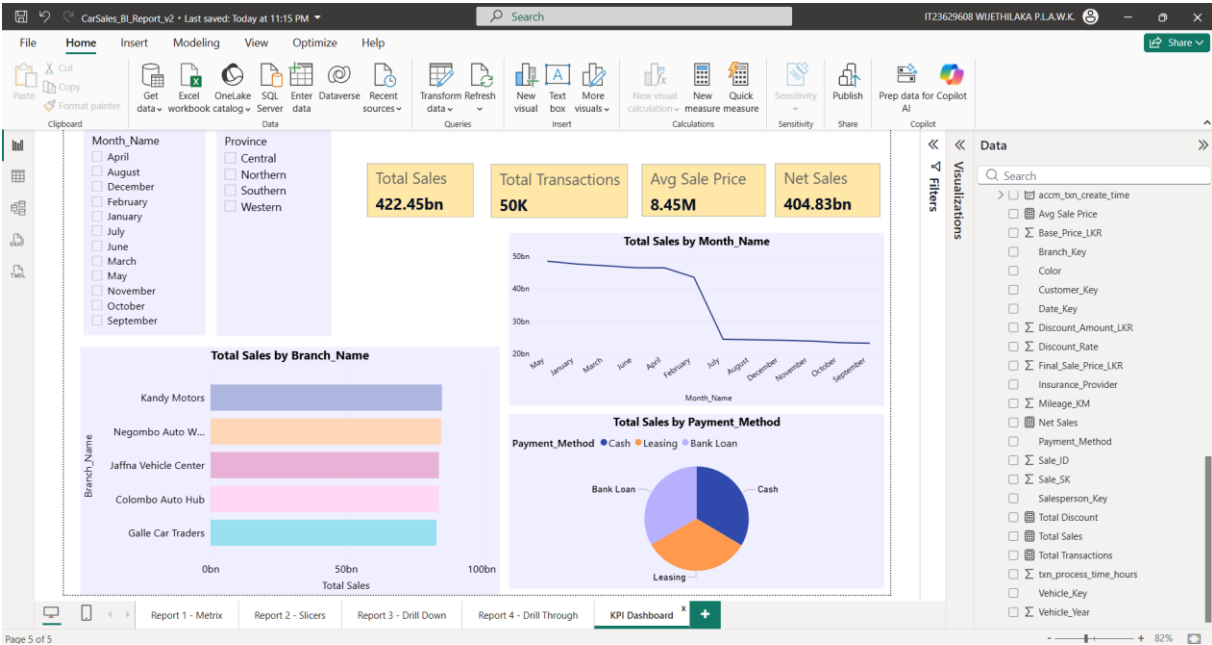


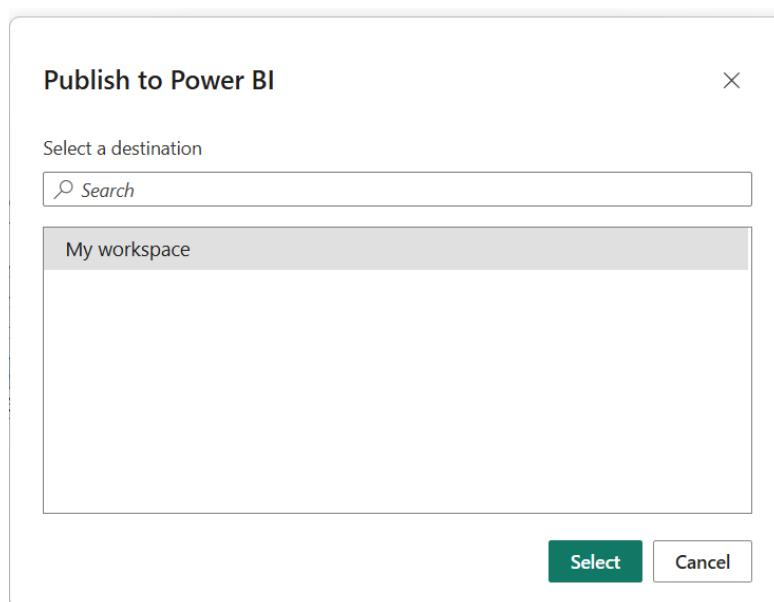
Figure 27: KPI Dashboard displaying overall car sales performance summary

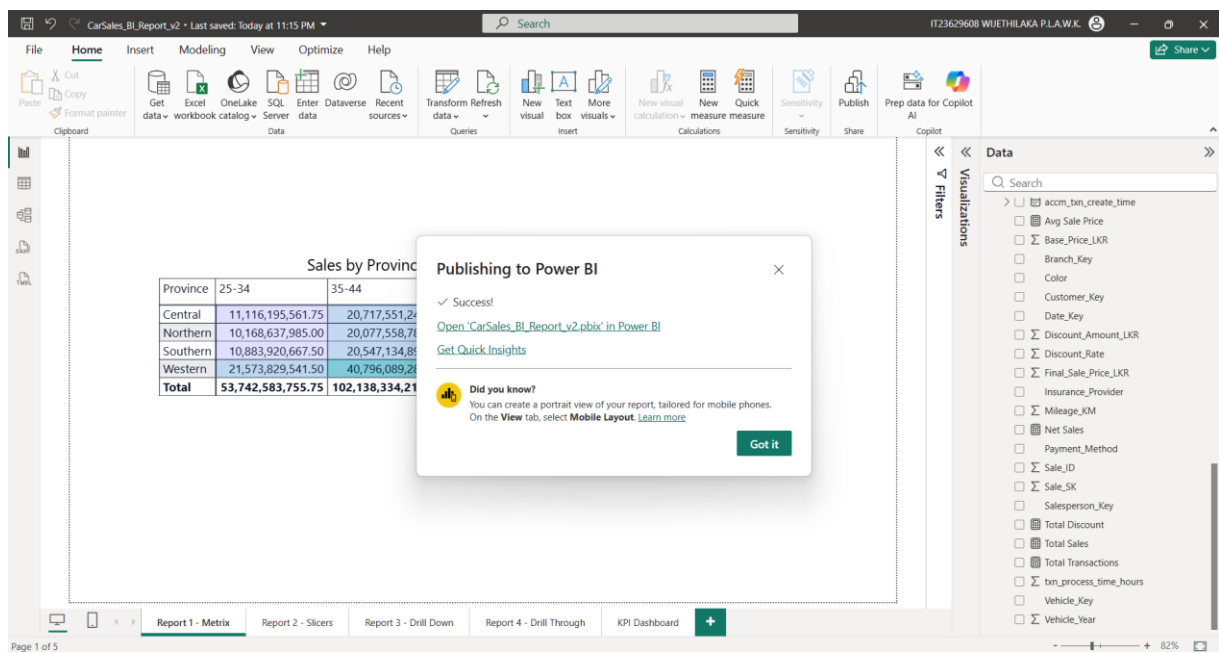
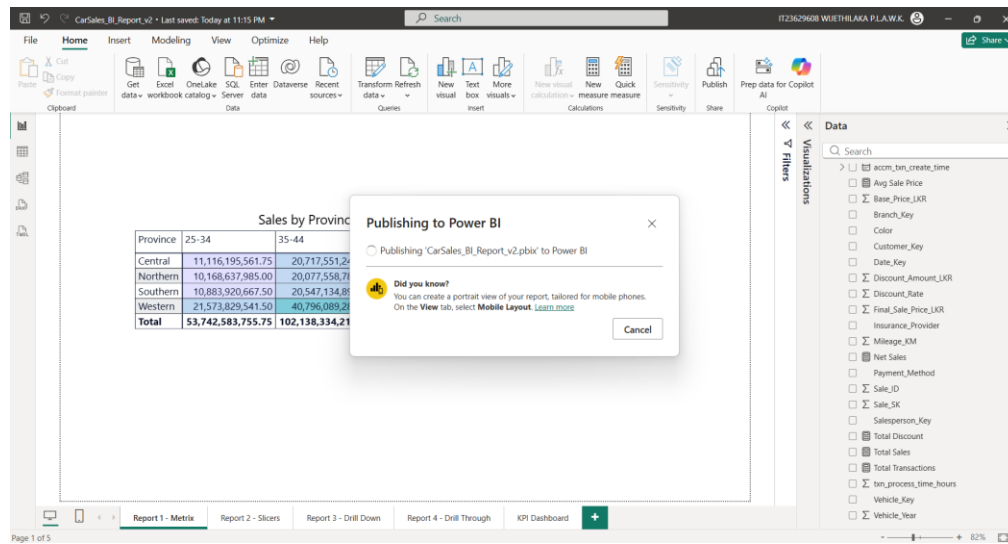
From the dashboard, **Kandy Motors** leads slightly in branch sales performance. The line chart reveals a notable **decline in sales from mid-year onwards**, and the donut chart shows that **Cash and Leasing** are the dominant payment methods compared to Bank Loan.

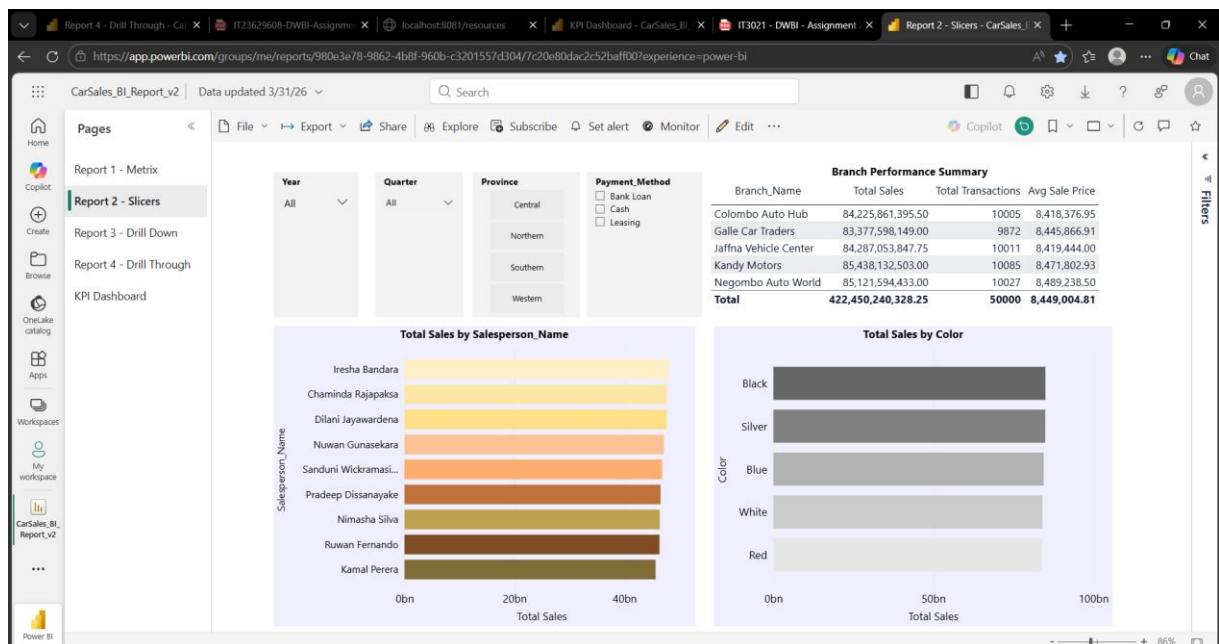
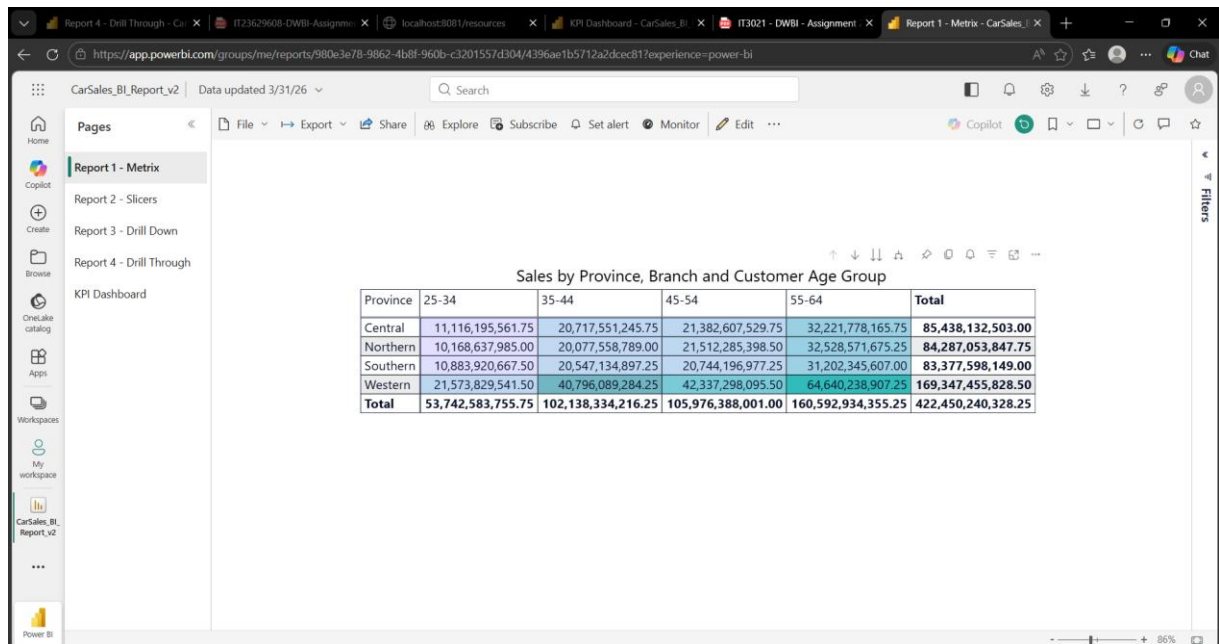
4.6 Publishing to Power BI Service

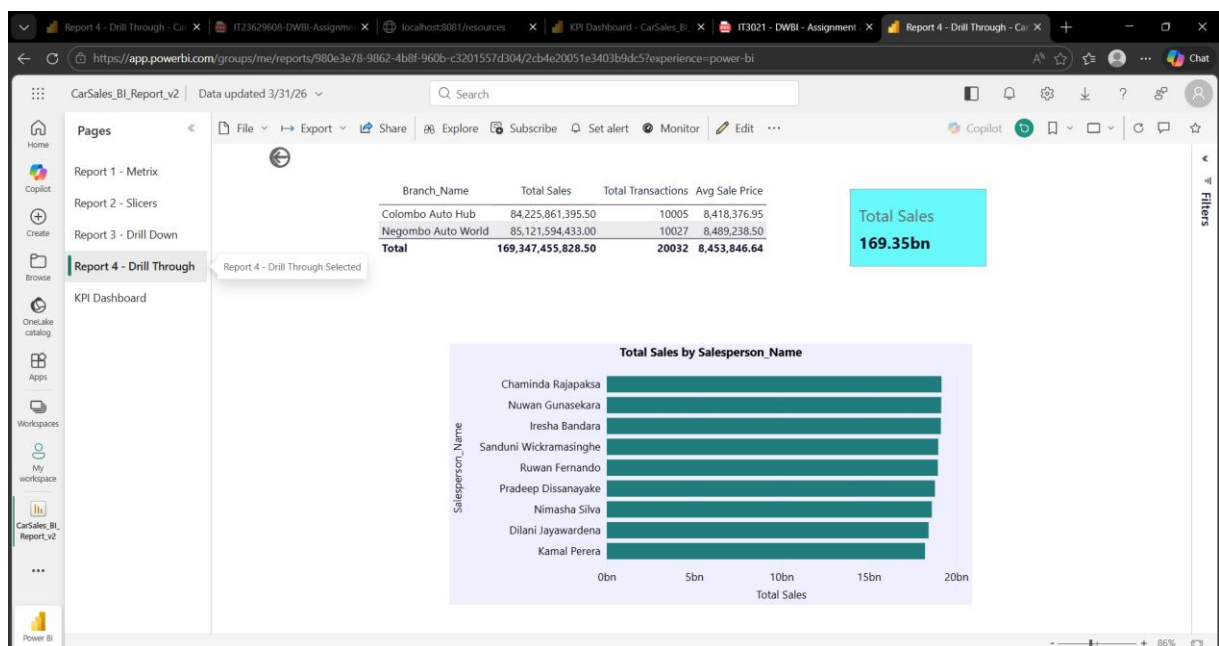
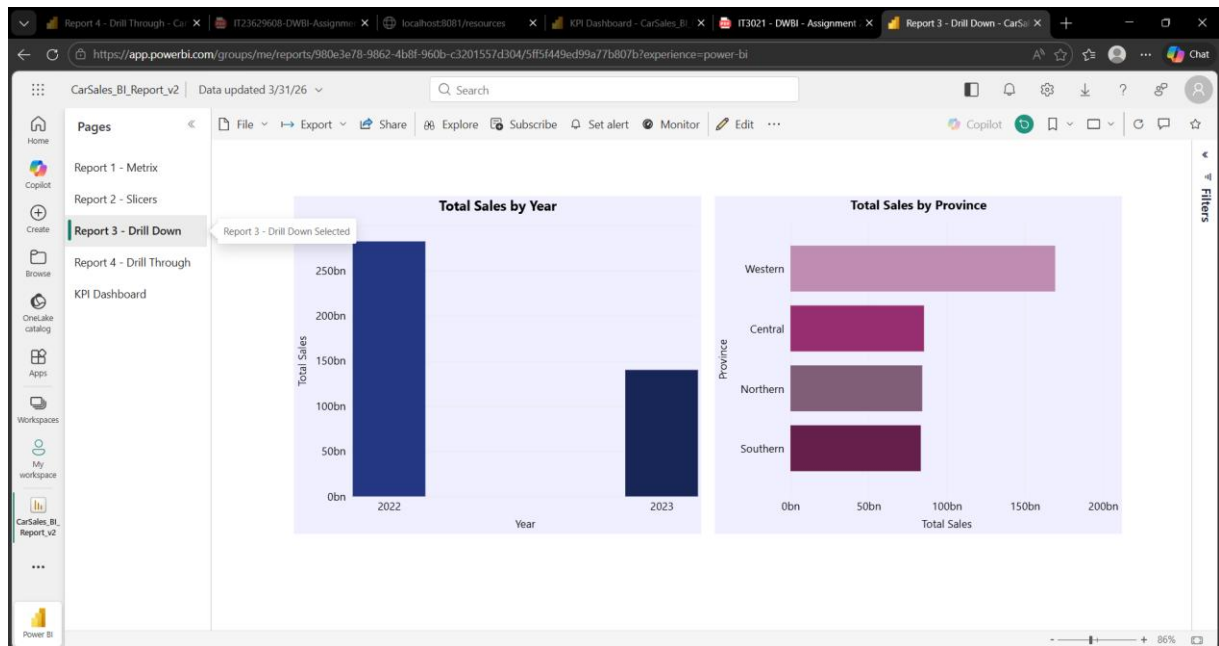
All developed reports have been successfully published to the Power BI Service (app.powerbi.com), allowing online access and demonstration. The publication of reports was carried out directly from Power BI Desktop, allowing reports to be uploaded to a specific workspace within the Power BI Service environment.

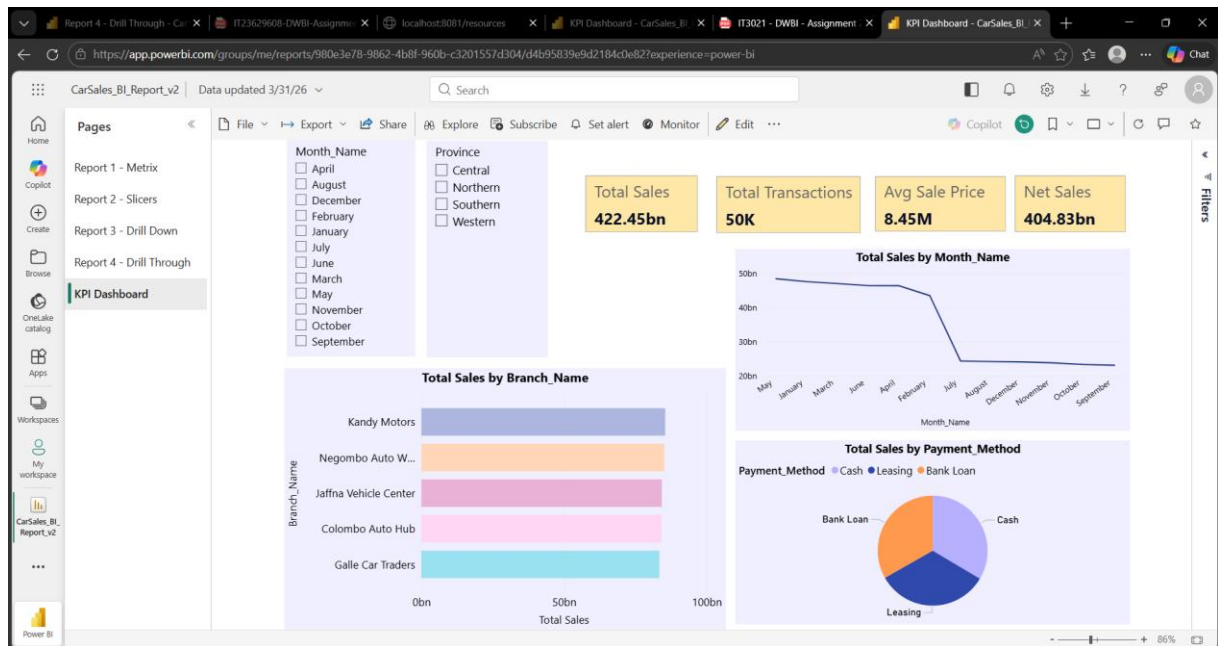
Reports have been accessed via a web browser, ensuring that reports are properly displayed and interactive. This step confirms that reports can be accessed and viewed remotely, allowing for better accessibility.











Step 5 – Conclusion

In conclusion, this project was successful in demonstrating the whole process of developing a data-driven business intelligence solution with the help of data warehousing and Power BI. The data warehousing solution was developed and implemented to manage huge structured data efficiently. The data integration process was also carried out with the help of ETL tools.

The implementation of SQL Server Analysis Services was useful in developing a data cube that facilitates faster data analysis. In addition to this, Power BI was used to create interactive reports such as matrices, slicers, drill-down reports, and drill-through reports that provide valuable information on sales. The developed solution is useful in making effective decisions by converting raw data into valuable information.